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Final Report

Version 2: formal model, algorithm comparison and a basis-adjusted lead-lag
agent

Learning When *Not* to Trade: Reinforcement Learning for
Cross-Venue Latency Arbitrage After Transaction Costs

Group X | Application Project

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AI-use disclosure: AI coding agents (an OpenAI Codex-based agent and Anthropic's Claude) implemented substantial parts of the software and drafted this report under the team's direction. The full statement follows the conclusion.

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Abstract

We asked whether a reinforcement learning (RL) agent can trade Bitcoin perpetual futures between Binance and Hyperliquid (HL) only when the price gap between the two venues pays for fees, spread and execution delay. Version 1 of this report documented two agents trained with proximal policy optimisation (PPO) that learned to stay flat. Version 2 states the task as a semi-Markov decision process (SMDP) with market dynamics that do not depend on the agent's actions, and uses that model to interpret the failure: version 1 observed the raw gap, which mixes a slow price offset between the venues (the basis) with a short-lived lead-lag component. On training days the raw gap had no gross edge at any threshold from 1 to 10 basis points (bps). A causal 60-second moving average of the gap, which serves as an approximate belief state for the basis, removed the offset; the remaining de-meaned gap earned 2.91 bps gross per trade at a 3 bps threshold on the same days. On this redesigned SMDP we compared PPO, a Double deep Q-network (Double DQN), fitted-Q iteration (FQI) that treats the exit as an optimal-stopping problem, and a contextual-bandit ablation without a learned exit; this comparison was registered after PPO's test results were known. All were selected on validation days and tested once in the unchanged simulator with fills 450 ms after each decision. On four test days that earlier attempts had already examined, PPO made \$76.91, \$37.91, \$2.21 and \$0.05 at taker fees of 0.5, 1, 2 and 3.5 bps per side, from 5,056 down to 45 trades; the last amount cannot be told apart from zero. Paired hour-block bootstrap intervals put PPO ahead of Double DQN, FQI and the bandit at 0.5 and 1 bps, level with the best hand-written rule (tuned on validation) at 0.5 bps and ahead of it at 1 bps (\$3.12, 90% interval \$0.80 to \$5.34). Double DQN beat PPO at 2 bps with 240 trades against 628. When fills were priced from HL's own clock instead of the received book, profits survived only at 0.5 bps (PPO \$27.05), and even there the bandit lost money. These are simulated results on reused test days, and they do not establish a deployable edge.

1 Introduction

1.1 Problem

Bitcoin perpetual futures (futures with no expiry date) trade on many venues at once, and a price move does not reach every venue's order book at the same moment. A trader who sees a move first on one venue can trade on a venue whose quotes have not yet adjusted and exit when the two prices converge. This is latency arbitrage. Budish, Cramton and Shim [1] describe the same mechanism in equity markets.

An order book lists resting buy orders (bids) and sell orders (asks). The mid price is the average of the best bid and the best ask, and the spread is their difference. We measure the gap as the Binance mid minus the HL mid, divided by the HL mid, in basis points (bps; 1 bps is 0.01%). An order that executes immediately against resting orders is a taker order: it buys at the ask, sells at the bid and pays a taker fee on every fill. "Per side" means per fill. A round trip therefore costs two fees, about one HL spread (median 0.12 bps) and any slippage, an execution price worse than the quoted one (0.1 bps per fill in our model). At 1 bps per side the round trip costs about 2.3 bps; at 3.5 bps per side about 7.3 bps.

The question was whether an RL agent can learn from recorded order books to enter only when the expected convergence covers these costs, exit at the right time, and do better than hand-written rules under identical costs on later days. The proposal named proximal policy optimisation (PPO) [2] as the main method and a deep Q-network (DQN) [3] as a value-based comparison.

1.2 What changed since version 1

Version 1 trained two PPO designs. Both ended with zero validation trades, and the selected policy of the first made zero trades on the test days, while a fixed 10 bps threshold rule made \$1.19 over 16 trades at 3.5 bps per side. Version 2 changed the problem formulation, the signal, the evaluation protocol and the set of algorithms:

- The task is written as an SMDP with a factored state, state-dependent action sets and exogenous market dynamics (Section 4). Version 1 is recast as a partially observable problem with a hidden basis (Section 4.6).
- The signal is the gap minus a causal estimate of the basis. The agent is consulted only at seconds where this de-measured gap is at least 2 bps and at every second while it holds a position, and the fee is part of the state, so one network serves every fee level.
- The primary fill delay is 450 ms after the decision instead of 150 ms, because HL's book reaches the recorder about 340 ms after HL's own timestamp (Section 3.2).
- A protocol registered before any validation result fixed the splits, the candidate runs and the selection rule. Five dated amendments followed; the one that made the 450 ms group primary and set the paired endpoint came after partial test results had been seen, and the one that added the new algorithms came after all PPO test results (Section 7).
- Double DQN [4], fitted-Q iteration [5, 6] and a contextual-bandit ablation (a one-step learner that only decides whether to enter) were trained on the same SMDP and compared with PPO and with hand-written rules (Section 6).

1.3 Contributions

1. A formal SMDP model of the task in which the market evolves independently of a 0.001 BTC order. Under that assumption, and given the simulator's fill model, any policy's return on recorded data follows by replay without policy-induced distribution shift, the exit is an optimal-stopping problem, and the entry is close to a contextual bandit. These consequences shaped the choice of algorithms.
2. A diagnosis of version 1's abstention that is consistent with the training data: on training days, entering on the raw gap lost money before fees at every threshold from 1 to 10 bps, so abstaining was a reasonable response to that observation. We did not run an ablation that isolates this cause.
3. A version 2 agent that trades. On the four test days, which were examined before, PPO's net profit after modelled costs was positive at 0.5, 1 and 2 bps per side and could not be told apart from zero at 3.5 bps, with fills 450 ms after the decision. Under the stricter HL-clock fill convention it was positive only at 0.5 bps per side.
4. A comparison of four learners, registered before the three new ones were trained but after the PPO test results were known. PPO was conclusively ahead of Double DQN, FQI and the bandit at 0.5 and 1 bps per side, tied the best hand-written rule at 0.5 bps and beat it at 1 bps. Double DQN beat PPO at 2 bps.
5. An analysis of sub-second data that estimates, conditional on the HL-clock convention, how much faster feeds would add, with numbers for each source of delay.

2 Data

The team recorded both venues' public order-book streams into one ClickHouse table: Binance btcusdt@depth20@100ms (20 levels per side, 9.32 snapshots per second on average) and HL 12Book (5 levels per side, 1.84 snapshots per second). The research range is the 30 complete

days from 3 September to 2 October 2026 (UTC), 28,917,283 raw rows, split by date into 22 training, 4 validation and 4 test days (Table 1). We call the test days of 29 September to 2 October test A; the day of 3 October, prepared separately, is test B. The data are public market feeds and contain no personal information.

Decisions are made on whole UTC seconds. Every row takes, for each venue, the latest snapshot received at or before its timestamp, so no row uses information that arrived later. Version 2 adds execution rows at 150, 300, 450 and 600 ms after each decision second (version 1 had 150 and 300 ms). Five-minute windows with any invalid row (stale or missing quotes, crossed books) are dropped whole; the extra offsets add quality checks, so 8,227 windows are kept instead of version 1’s 8,245: 5,970 training, 1,117 validation and 1,140 test windows, plus 285 on 3 October. At decision rows the median HL quote age (row time minus HL’s exchange timestamp) was 623 to 640 ms; the median top-of-book spread was 0.012 bps on Binance and 0.12 bps on HL.

Table 1: Splits. Window counts are version 2’s. Gap statistics are version 1’s, computed on its slightly larger set of windows (5,984, 1,121, 1,140 and 286); gap is $10^4 \times (\text{Binance mid} - \text{HL mid})/\text{HL mid}$.

Split	Dates (2026)	Windows (version 2)	gap (bps)		Seconds with gap $\geq$ 7 bps
			Median	p99	
Train	3–24 Sep	5,970	1.34	12.57	18.04%
Validation	25–28 Sep	1,117	1.30	5.20	0.084%
Test A	29 Sep–2 Oct	1,140	1.04	4.86	0.097%
Test B	3 Oct	285	0.61	2.31	0.0047%

The training split has far more large gaps than the later splits. From 18 to 23 September the Binance mid sat below the HL mid for whole days, with daily median signed gaps between -2.4 and -9.8 bps, and these six days held 99.7% of the training seconds with an absolute gap of 7 bps or more. On those days the large gaps came from a slow, persistent offset between the venues (a basis). Perpetual futures are tied to the spot price by funding payments between long and short holders, and a funding difference between venues can sustain such an offset; funding rates are not in the source table, so we could not test this. This offset is why the raw gap had no edge on training days (Section 5).

3 Environment and execution model

3.1 Replay and costs

Test results for the frozen choices in the main tables come from version 1’s simulator, unchanged; it follows the Gymnasium interface [7]. An order decided at second t fills at the replay row L ms later. A fill walks the far side of the recorded HL book for the order size and pays the volume-weighted average price (VWAP) over the levels it consumes, moved 0.1 bps against the trader as slippage; a taker fee is charged on the fill notional (price times quantity). An entry is rejected if HL’s quote age or receive age (row time minus the time the recorder received the book) exceeds 1,000 ms at the decision or at the fill. Positions are 0.001 BTC (about \$84), at most one at a time, closed by force 30 s after the entry fill or at the window’s last row. Table 2 lists the settings.

The simulator steps one second at a time and runs at about 3,500 steps per second, too slow for many passes of PPO over 22 days. Training, validation selection, the stress tests, the HL-clock

fills and the same-snapshot split of Section 9 therefore use a fast replay over a precomputed per-second table that runs about 1,500 PPO steps per second per process. Unit tests check that it reproduces the simulator trade for trade for rule policies at all four latencies, including forced exits, and the selected PPO policies gave the same trade counts and the same test P&L to the cent in both (Tables 6 and 17). The HL-clock convention has no counterpart in the simulator, so no such check covers it.

Table 2: Execution settings shared by every policy in version 2.

Parameter	Value
Taker fee per side, evaluation	0.5, 1, 2, 3.5 bps (4.5 bps stress)
Taker fee per side, training (sampled per window)	0.5, 1.0, . . . , 3.5 bps
Fill latency after the decision second	450 ms; 150 ms (sensitivity); 300, 600 ms (stress)
Slippage per fill	0.1 bps
Spread multiplier	1.0 (1.5 as a stress test)
Position size	0.001 BTC, one position at a time
Maximum holding time	30 s from the entry fill
Entry freshness limit (quote and receive age)	1,000 ms
Reference capital	\$10,000, reset each window
Funding	not available; set to zero

3.2 Fill-timing conventions and why they matter

The simulator prices a fill at $t + L$ from the latest HL book *received* by the recorder at or before $t + L$. Two facts make this optimistic. First, HL’s book arrives late: on six training days the receive time minus HL’s exchange timestamp had a median of 338 ms (10th to 90th percentile 304 to 429 ms; this includes any clock offset between the machines). Second, HL publishes a new book only every 540 ms at the median. A fill priced at $t + 150$ ms from the received book therefore often uses the same snapshot the decision already saw, a price that was known before the order was sent: 71% of the frozen 450 ms PPO policy’s test entries did so when replayed with 150 ms fills (Table 13).

We therefore used three conventions. *Received book* is the simulator’s. *HL clock* prices the fill from the last HL book whose exchange timestamp is at or before $t + L$, which assumes our clock matches HL’s. *Next snapshot*, used only in the sub-second analysis, takes the first HL book stamped after $t + L$. On the six training days, entering at a de-meaned gap of 3 bps or more and exiting 3 s later earned 3.39 bps gross per trade with received-book fills at 150 ms, 2.04 bps with HL-clock fills and 0.63 bps with next-snapshot fills. A received-book fill at 450 ms approximates a 150 ms order delay plus HL’s publication lag, so the 450 ms group is primary and HL-clock fills are a stress test. The HL-clock stress at $L = 450$ ms assumes that the order reaches HL 450 ms after the decision on HL’s own clock. It therefore combines a 450 ms order path with HL’s publication lag and is roughly a received-book fill 790 ms after the decision (450 plus the 338 ms median lag). The HL-clock convention that matches a 150 ms order delay is HL clock at 150 ms, whose 2.04 bps on the six days is close to the 2.05 bps that received-book fills at 450 ms earned on all 22 training days (Section 5).

4 Formal problem: a semi-Markov decision process

4.1 Episodes and decision epochs

Each retained five-minute window is one finite-horizon episode with whole seconds $t \in \{0, \dots, 299\}$. The agent is consulted only at decision epochs $0 \leq t_0 < t_1 < \dots < t_K \leq 299$, a subset of seconds that depends on the market path and on the agent's position. Between epochs the position stays as it is. Because the sojourn time $\tau_k = t_{k+1} - t_k$ varies, the process is a semi-Markov decision process: in the options framework of Sutton, Precup and Singh [8], “stay as you are until the next epoch” is a fixed option whose duration depends on the state. Figure 1 shows the two position phases.

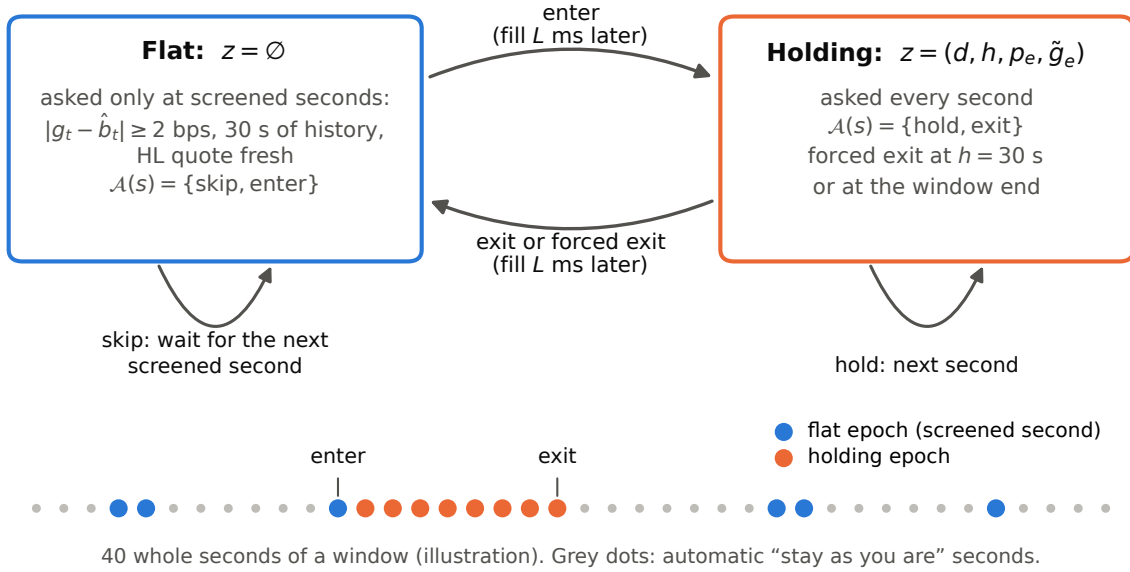


Figure 1: The decision process. A flat agent is asked only at screened seconds; a holding agent is asked every second. Orders fill L ms after the decision. The timeline below is an illustration of how epochs fall within a window, not data.

4.2 State

The state at epoch k factorises as

$$s_k = (x_{t_k}, z_k, f). \quad (1)$$

The *market state* x_t holds what the recorded books show at second t : the gap $g_t = 10^4(m_t^B - m_t^H)/m_t^H$, where m_t^B and m_t^H are the Binance and HL mids; the basis estimate $\hat{b}_t$ and the de-measured gap $\tilde{g}_t = g_t - \hat{b}_t$; 1 s and 5 s mid returns on both venues; the 30 s realised volatility (standard deviation) of Binance's 1 s returns; the HL spread; HL's quote age and receive age; the order-book imbalance on each venue, $(\text{bid volume} - \text{ask volume})/(\text{total volume})$ over the displayed levels; the time left in the window; and the length of uninterrupted history n_t . The basis estimate is an exponential moving average (EMA) of earlier gaps,

$$\hat{b}_t = (1 - \alpha)\hat{b}_{t-1} + \alpha g_{t-1}, \quad \alpha = \frac{2}{61}, \quad (2)$$

a 60-second span that excludes the current second and restarts wherever retained windows are not adjacent.

The *position state* is $z = \emptyset$ when flat and $z = (d, h, p_e, \tilde{g}_e)$ when holding, with direction $d \in \{-1, +1\}$ (+1 is long HL), holding time h in seconds, entry fill price p_e and the de-measured gap

at entry $\tilde{g}_e$. The fee f per side is fixed within an episode and drawn per training window from $\mathcal{F}_{\text{train}} = \{0.5, 1.0, \dots, 3.5\}$ bps. The agent observes f , so the task is an ordinary MDP whose state is augmented with an episode-constant fee, and one policy $\pi(a \mid x, z, f)$ serves every fee. This differs from the contextual MDPs of Hallak et al. [9], where the context is hidden.

The network sees a 20-dimensional observation $o = \psi(x, z, f)$ of the state (1): an in-position flag, the direction-signed market features, the fee, holding time, unrealised profit at the decision row's exit price, $\tilde{g}_e$, time left and history length. Signed features are multiplied by the trade direction ($\text{sign } \tilde{g}_t$ when flat, d when holding), so the network does not have to learn the long and short cases separately. Every feature is divided by a fixed constant and clipped to $[-10, 10]$; no scale is fitted to any split.

4.3 Actions

The action set depends on the state:

$$\mathcal{A}(s) = \begin{cases} \{\text{skip, enter}\} & z = \emptyset, \\ \{\text{hold, exit}\} & z \neq \emptyset. \end{cases} \quad (3)$$

“Enter” opens a 0.001 BTC HL position in direction $\text{sign } \tilde{g}_t$: long HL when Binance is above HL after the basis is removed. A flat agent is consulted only when the screen

$$C(x_t) = \mathbf{1}\{|\tilde{g}_t| \geq 2 \text{ bps}\} \cdot \mathbf{1}\{n_t \geq 30 \text{ s}\} \cdot \mathbf{1}\{\text{HL quote and receive age} \leq 1,000 \text{ ms}\} \cdot \mathbf{1}\{t + 2L < T\} \quad (4)$$

equals one, where T is the time of the window's last row, so that an entry would fill at least L before the window ends; a holding agent is consulted every second. Restricting the epochs and the action sets (3) in this way is action masking written into the model: the seconds that version 1 asked about, where the average entry loses money after fees, are no longer decisions. Both phases reuse the same two output indices, and the in-position flag tells the network which phase it is in.

4.4 Transitions

Orders fill at $t + L$, between two whole seconds. We let the market state x_{t+1} also carry the recorded HL execution prices at $t + L$ (the per-second table stores the book 150, 300, 450 and 600 ms after each decision second), so that every fill caused by an action at t is a function of x_{t+1} . The transition then factorises into an exogenous market part and a deterministic position part:

$$P(x_{t+1} \mid x_t, z_t, a_t) = P(x_{t+1} \mid x_t), \quad (5)$$

$$z_{t+1} = \Phi(z_t, a_t, x_{t+1}). \quad (6)$$

Equation (5) assumes that a 0.001 BTC order does not move either venue's book. Φ applies the fill rules of Section 3. An entry fills at $t + L$ at the depth VWAP plus slippage, or is rejected if a quote is stale or the displayed depth does not cover the order. An exit fills at $t + L$ under the same depth condition; if it is rejected, the position stays open. The position is closed by force at $h = 30 \text{ s}$ or at the window's end. The fee does not change within the window. Because the next epoch time and the next position are deterministic functions of the current position, the action and the market path, the SMDP kernel for a sojourn of τ seconds from epoch time $t = t_k$ is

$$P(s', \tau \mid s_k, a_k) = \int \mathbf{1}\{(z', \tau) = \Psi(z_k, a_k, x_{t:t+\tau})\} \prod_{u=t}^{t+\tau-1} P(x_{u+1} \mid x_u) dx_{t+1} \cdots dx_{t+\tau-1}, \quad (7)$$

where $s' = (x', z', f)$, the endpoint $x_{t+\tau}$ is fixed to x' , and the integral runs only over the interior path $x_{t+1}, \dots, x_{t+\tau-1}$ (it is empty when $\tau = 1$). The map Ψ applies Φ of Equation (6) second by second and returns the first later second at which the agent is consulted again, by the screen when flat or at once when holding, together with the position at that second. The Markov property of x itself is an approximation: x is a summary of two order books, not the full market state.

4.5 Reward and objective

Equity is cash plus position times the HL mid, $E_t = c_t + q_t m_t^H$, where the position q_t is $-\bar{q}$, 0 or $\bar{q}$ with the fixed order size $\bar{q} = 0.001$ BTC. A fill at price p changes cash by $-\Delta q p - f \cdot 10^{-4} |\Delta q| p$. The reward is the change in marked equity between epochs, in basis points of the window's starting notional,

$$r_k = \frac{E_{t_{k+1}} - E_{t_k}}{\bar{q} p_0 \cdot 10^{-4}}, \quad k = 0, \dots, K, \quad \sum_{k=0}^K r_k = \frac{E_{\text{end}} - E_{\text{start}}}{\bar{q} p_0 \cdot 10^{-4}}, \quad (8)$$

where p_0 is the HL mid at the start of the window and t_{K+1} denotes the window's end, where equity equals cash. The sum telescopes to the window's net profit and loss (P&L) because the agent is flat before t_0 and every position is closed by the end; a unit test checks this. There is no trade bonus. For one trade the net payoff in bps of entry notional is

$$\Pi = d \frac{p_x - p_e}{p_e} \cdot 10^4 - f - f \frac{p_x}{p_e}, \quad (9)$$

with entry and exit fill prices p_e and p_x that already include the spread crossing and slippage. The training objective is

$$J(\pi) = \mathbb{E}_{f \sim \mathcal{U}(\mathcal{F}_{\text{train}})} \mathbb{E}_{\pi} \left[\sum_{k=0}^K \gamma^k r_k \mid f \right]. \quad (10)$$

PPO and Double DQN use $\gamma = 0.99$ per epoch. The textbook SMDP discount would be γ^{τ_k} per epoch [8]; discounting per epoch instead treats a long wait between screened seconds like one second. In Equation (10) the exponent k counts epochs from the start of the window, and a window contains dozens of epochs (on test A at 0.5 bps, PPO held a position for about 48 seconds per window, each a holding epoch), so rewards late in a window are weighted by about $0.99^{48} \approx 0.62$ or less. The learners do not follow this weighting literally. PPO's and DQN's updates use discounted returns from each epoch onward and do not down-weight late epochs, so in practice γ sets how far each decision looks ahead. Within a trade of 5 to 20 holding epochs the discount factor falls to between 0.95 and 0.82, so this lookahead is slightly more conservative than the undiscounted payoff of the trade. FQI uses $\gamma = 1$ within a trade. All reported results are undiscounted sums of net P&L in US dollars over test windows.

4.6 Version 1 as a POMDP with a latent basis

Write the gap as a slow basis plus a transient lead-lag term,

$$g_t = b_t + u_t, \quad (11)$$

where b_t drifts over minutes to days and u_t is the part that HL closes within seconds. A trade's payoff depends on how much of u_t converges before the exit; b_t persists through the 30 s holding limit and contributes nothing. Version 1 observed g_t and not b_t , so it was a partially observable

MDP (POMDP) [10]: the same observation $g_t = 8$ bps can mean $u_t = 8$ (a trade worth taking) or $b_t = 8$ (a trade that pays fees and never converges). The optimal policy of a POMDP depends on the belief $\beta_t = P(b_t | o_{0:t})$, and a policy that sees only the current observation cannot represent it. On training days, entering on a raw gap of 1 to 10 bps lost money on average before fees, with the fill 150 ms after the decision and the exit 3 s later; at 15 bps the small positive gross edge (0.63 bps) did not cover the round-trip fee even at 0.5 bps per side (Section 5). This measurement uses one fixed exit, so it does not establish the best policy for that observation, but it suggests that on the training distribution the best policy given the raw gap alone is close to always flat.

Version 2 adds $\hat{b}_t$ to the state. Suppose b_t follows a Gaussian random walk and u_t is independent Gaussian noise around it (a local-level model). Then the minimum mean-squared-error estimate of b_t from past gaps is an exponentially weighted average of those gaps [11], the steady state of the Kalman filter for that model; without the Gaussian assumption it is the best linear estimate. In the steady state the posterior variance is constant, so the posterior mean summarises the belief. Equation (2) is such a filter, so $\hat{b}_t$ serves as the belief state and $\tilde{g}_t = g_t - \hat{b}_t$ estimates u_t . With it the observation is approximately Markov with respect to the quantity that determines the payoff. The data depart from the model in three ways. The gain is fixed, not fitted. The lead-lag term is correlated from one second to the next, not white noise. And the EMA restarts from the current gap at the start of each run of adjacent windows, so at the earliest screened second (30 s of history) it still puts about 38% of its weight, $(1 - \alpha)^{29}$, on the first gap of the run. The filter also lags a sudden change in the basis by tens of seconds.

4.7 Consequences of exogenous market dynamics

Equation (5) has three consequences.

4.7.1 Counterfactual evaluation by replay. Given the fill model and no market impact, the return of any policy on a recorded window is a deterministic function of the policy and the recorded path, computed by replay. Off-policy evaluation therefore has no policy-induced distribution shift and needs no importance weights. Uncertainty remains from the fill convention (Section 9), from the assumptions of no impact and no competing traders, from which windows and days were sampled, and from changes in the market between days. Replay also makes paired comparisons possible: two policies replayed on the same windows face the same market path, so their per-window difference removes the market noise the two share.

4.7.2 The exit is an optimal-stopping problem. While holding, the agent chooses between stopping (exit) and continuing (hold). Let $G(s)$ be the payoff (9) of exiting now, which depends on the fill at $t + L$ and is therefore random at decision time, and let F be the payoff of the forced exit at $h = 30$ s. After an exit the agent is flat and may enter again later in the window, so the exact value of exiting is $\mathbb{E}[G(s) + V_\emptyset(s_{\text{after}}) | s]$, where $V_\emptyset \geq 0$ is the value of being flat at the next screened second s_{after} . Treating each trade as a separate stopping problem sets $V_\emptyset$ to zero. That is a second approximation, alongside the greedy entry below: it ignores the value of being free to trade again, lowers the value of exiting and so biases the exit toward holding. With it, the Bellman equations of the stopping problem are

$$Q(s, \text{exit}) = \mathbb{E}[G(s) | s], \quad (12)$$

$$Q(s, \text{hold}) = \mathbb{E}[V(s') | s, \text{hold}], \quad V(s) = \max_{a \in \mathcal{A}(s)} Q(s, a), \quad (13)$$

with $V(s) = F$ at $h = 30$ s. Because the market path does not depend on the decision, every recorded holding path supplies the payoff of exiting at every second, so the continuation values

can be fitted by regression on recorded paths without exploration. This is the idea behind regression-based pricing of American options (options that can be exercised at any time before expiry), such as the method of Longstaff and Schwartz [12], with recorded paths in place of simulated ones. Section 6.3 describes how our implementation differs from theirs.

4.7.3 The entry is close to a contextual bandit. For a flat agent,

$$Q(s, \text{enter}) = \mathbb{E}[\Pi + V_{\emptyset}(s_{\text{after}}) \mid s], \quad Q(s, \text{skip}) = \mathbb{E}[V_{\emptyset}(s_{\text{next}}) \mid s], \quad (14)$$

where Π is the trade payoff under the exit policy and $V_{\emptyset}$ is the value of being flat later in the window, as in the exact exit value above. If the two continuation values were equal, the agent should enter exactly when $\mathbb{E}[\Pi \mid x, f] > 0$: a contextual bandit with context (x, f) , reward Π for entering and 0 for skipping. They are not equal for two reasons. Holding blocks later entries (screened seconds pass while the agent is busy), and skipping keeps the option to enter later at a larger de-meaned gap. Both effects vanish when screened seconds are far apart and grow when they come in runs.

5 Why the first two PPO attempts abstained

Version 1 ran two PPO designs on the raw gap. Attempt 1 decided every second among HOLD, LONG, SHORT and EXIT. It trained four candidates for about one million steps each (seeds 7, 17, 27 and a retry with five times the entropy coefficient) with a 3.5 bps fee. Every candidate cut the share of entry actions below 1% within 176,128 to 229,376 steps, and all four made zero validation trades. The selected policy made zero test trades, while a 10 bps threshold rule made \$1.19 over 16 test trades. Attempt 2 asked skip or enter at seconds with an absolute gap of at least 6 bps, warm-started the policy by supervised learning on counterfactual payoffs (1.35% of 317,232 training labels were positive at 4.5 bps per side), and fine-tuned with PPO. All three PPO seeds made zero validation trades. Version 1 listed several properties of the problem that were consistent with this outcome without testing which caused it.

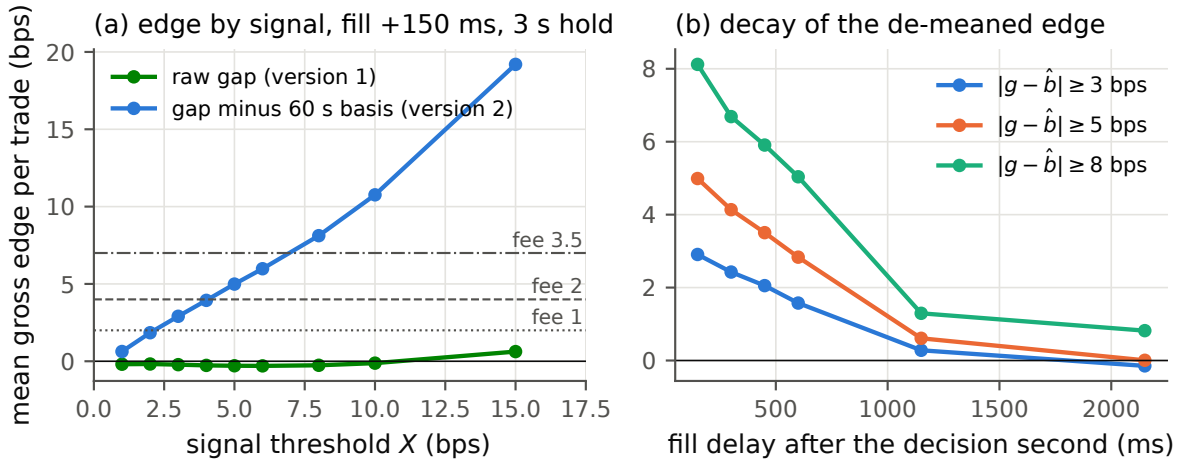


Figure 2: Training days only (3–24 September). Every decision second with $|\text{signal}| \geq X$ enters 0.001 BTC on HL in the signal direction at the recorded book 150 ms later (depth VWAP plus 0.1 bps slippage) and exits 3 s later the same way. Gross edge includes spread and slippage but not fees; a trade breaks even when gross edge equals twice the fee per side (horizontal lines, labelled by fee). (a) Raw gap against the gap minus its 60 s EMA basis. (b) The de-meaned edge against the fill delay. From `analysis/train_signal_analysis.json`.

Figure 2 shows the gross edge of each signal on the training days. A trade triggered by the raw gap lost between 0.12 and 0.30 bps gross on average at every threshold from 1 to 10 bps, before any fee; only the 3,098 signals at 15 bps or more over 22 days averaged a positive 0.63 bps, less than the two fees of a round trip even at 0.5 bps per side. The de-meaned gap earned 1.84 bps at a 2 bps threshold, 2.91 bps at 3 bps, 4.99 bps at 5 bps and 8.12 bps at 8 bps, and at a 3 bps threshold with a 1 bps fee per side its net was positive on all 22 training days. The edge decays quickly after the signal: at the 3 bps threshold it was 2.91 bps with the fill at 150 ms, 2.05 bps at 450 ms, 1.58 bps at 600 ms and 0.28 bps at 1,150 ms.

In the terms of Equation (11), version 1’s observation aliased the basis b_t with the lead-lag term u_t . For an agent that sees only the raw gap on these training days, entering with a 3 s exit lost money at every measured threshold once fees are paid. One fixed exit does not show that the best memoryless policy is always flat; a raw-gap rule that exits when the gap closes made money at low fees on the later days (Table 6). PPO converged to near-abstention, which is consistent with the measurement. We attribute the failure mainly to the observation; we did not train another optimiser on version 1’s observation. Version 2 also changed the decision epochs, made the fee an input and let the agent learn the exit, so the test results below do not isolate the basis feature. Two pieces of evidence still favour the basis explanation. On the test days a fixed-hold rule on the de-meaned gap beat the version 1 raw-gap rule at 0.5 and 1 bps (Table 6). Also, on the six basis days, 18 to 23 September, between 21.9% and 98.2% of seconds had an absolute raw gap of 7 bps or more, while only 0.011% to 0.14% had an absolute de-meaned gap that large, the same range as on the other 16 training days (0.002% to 0.19%). The EMA removed the offset that dominated version 1’s training data.

6 Algorithms

All four learners used the same 22 training days, decision epochs, 20-feature observation, fee sampling and validation selection rule. PPO was trained on the table built before the amendment-3 fix, which differs from the rebuilt table used by the other three in 389 of about 14.3 million fill cells (Section 13). Table 3 summarises what each learner assumes and which question it answers. Appendix B lists every setting.

6.1 PPO

PPO [2] updates the policy by maximising the clipped surrogate objective

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_k \left[\min(\rho_k(\theta) \hat{A}_k, \text{clip}(\rho_k(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_k) \right], \quad \rho_k(\theta) = \frac{\pi_\theta(a_k | o_k)}{\pi_{\theta_{\text{old}}}(a_k | o_k)}, \quad (15)$$

with $\varepsilon = 0.2$ and advantages $\hat{A}_k$ from generalised advantage estimation [13] ($\gamma = 0.99$, $\lambda = 0.95$), plus a value loss and an entropy bonus (coefficient 0.01, or 0.03 in half of the 150 ms runs). The clip in Equation (15) removes any gain from moving the probability ratio ρ_k outside $[1 - \varepsilon, 1 + \varepsilon]$, which keeps each update close to the policy that collected the data. We used Stable-Baselines3 [14] with 8 parallel environments and separate 64×64 tanh policy and value networks; each run trained for 2,002,944 steps with checkpoints every 250,000 steps. Table 14 in Appendix B lists the remaining settings.

6.2 Double DQN

DQN [3] learns $Q_\theta(o, a)$ from a replay buffer with a periodically copied target network θ^- . The standard target $r + \gamma \max_{a'} Q_{\theta^-}(o', a')$ takes a maximum over noisy estimates and is biased

Table 3: The four learners. Time is wall-clock training time per seed on one CPU process, from the saved run settings (450 ms runs).

Learner	How it learns	Question it answers	Time/seed
PPO	On-policy actor-critic; fresh rollouts from the fast replay; exploration by a stochastic policy	Whether the proposal’s main method learns entry and exit jointly once the state is right	1,591–1,613 s
Double DQN	Off-policy value learning; replay buffer; random actions with a decaying probability; Double-DQN target (Section 6.2)	Whether replay of past transitions and a value target do better than on-policy updates	1,171–1,176 s
FQI (stopping) + greedy entry	Batch regression on all counterfactual training paths; no exploration	How far a method that exploits the exogenous dynamics gets with little compute	95–104 s
Contextual bandit	One-step regression of the payoff of a fixed 3 s trade	What the learned exit adds	(in FQI run)

upwards. In this task the bias reaches both actions of a flat agent: $Q(\text{skip})$ through the maximum at the next flat epoch, and $Q(\text{enter})$ through the maxima at the holding epochs that follow. Entering is followed by a chain of holding epochs, each adding a maximum, so the overestimation may accumulate more on the enter branch and favour entering. This is a bias in the targets. Noise in the estimate of $Q(\text{enter})$ at the decision itself also leads to marginal entries, and the Double target does not address that. Double DQN [4] selects the next action with the online network and evaluates it with the target network:

$$y = r + \gamma(1 - \text{done}) Q_{\theta} \left(o', \arg \max_{a'} Q_{\theta}(o', a') \right). \quad (16)$$

We subclassed Stable-Baselines3’s DQN to use this target with a Huber loss (squared error for small errors, absolute error for large ones). Exploration was ε -greedy: a random action with probability ε , decayed linearly from 1.0 to 0.02 over the first 20% of training. Seeds 7, 17 and 27 each ran for 1,000,000 steps with a hard target copy every 5,000 steps and checkpoints every 250,000 steps; Table 14 lists the rest. Double DQN is the value-based comparison the proposal promised.

6.3 Fitted-Q iteration for the exit, with a greedy entry

FQI [5] turns Bellman equations, here those of the stopping problem (12)–(13), into a sequence of supervised regressions on a fixed batch of transitions; with a neural network as the regressor it is neural fitted Q iteration [6]. Exogeneity (Section 4.7) supplies the batch without any exploration. For every admissible training entry (a screened second where the entry would fill), the recorded table gives the holding path and the exact payoff (9) of exiting at each of the following seconds $h = 1, \dots, 29$, together with the forced-exit payoff F_i . Each of the 39,921 admissible entries was used twice with fees drawn from $\mathcal{F}_{\text{train}}$, giving 79,842 paths. With features $\phi_{i,h}$ of path i at holding second h , exit payoff $X_{i,h}$ and last decision second H_i , each

iteration n computes

$$V_{i,h}^{(n)} = \max_{a \in \mathcal{A}_{i,h}} Q_a^{(n)}(\phi_{i,h}), \quad y_{i,h}^{\text{hold}} = \begin{cases} V_{i,h+1}^{(n)} & h < H_i, \\ F_i & h = H_i, \end{cases} \quad y_{i,h}^{\text{exit}} = X_{i,h}, \quad (17)$$

where $\mathcal{A}_{i,h}$ contains exit only if the book at that second can fill it, and fits a two-headed network to these targets (the exit head only where an exit could fill). The network is a $20 \rightarrow 64 \rightarrow 64 \rightarrow 2$ tanh perceptron trained with Adam at learning rate 10^{-3} and minibatches of 4,096. The loss is a Huber loss with a 2 bps threshold, so each head estimates a robust central value of its targets (an M-estimate) rather than their mean. We ran 15 iterations with one pass over the data each and kept the network between iterations. The exit policy is to exit at the first second where $Q_{\text{exit}} \geq Q_{\text{hold}}$.

This is fitted value iteration of the stopping problem, in the spirit of regression-based American option pricing, but it differs from the Longstaff–Schwartz method [12] in its targets. Longstaff and Schwartz work backwards one date at a time and regress the realised payoff of the current exercise rule, using fitted values only to decide when to exercise; this avoids the upward bias of taking a maximum over fitted values. Our targets are maxima of the fitted network at $h + 1$, updated for all h at once, as in the value-iteration variant of Tsitsiklis and Van Roy [15].

For the entry, the realised payoff Π_i of each path under the learned exit was regressed on the flat observation (fee included) for 8 epochs with the same Huber loss, and the agent enters when the prediction is positive. This approximates the bandit rule of Equation (14), enter when $\mathbb{E}[\Pi \mid x, f] > 0$, with a robust central value in place of the mean; it ignores the option of waiting for a larger gap and the cost of being busy. Training took 95 to 104 s per seed, about one sixteenth of PPO’s time.

6.4 Contextual-bandit ablation

The bandit uses the same entry regression with every trade closed at the first fillable second at least 3 s after entry, the horizon of the training-day analysis. There is no sequential decision after entry. Comparing it with FQI, which shares the entry method, indicates what the learned exit adds; because the two entry regressions are fitted to different payoffs, the two also enter on different seconds. Amendment 5 registered the bandit’s exit as the validation-tuned convergence rule at each fee; the implementation used the fixed 3 s exit instead. We report the bandit as implemented and treat this as a deviation from the protocol.

6.5 Hand-written rules

All rules ran in the same simulator with the same costs and windows. The version 1 raw-gap rule enters at $|g_t| \geq X$ and exits when the directional raw gap is back to 0.5 bps. The fixed-hold de-meaned rule uses the agent’s screen, enters at $|\tilde{g}_t| \geq X$ and exits after H seconds. The convergence-exit de-meaned rule (added by amendment 2) enters at $|\tilde{g}_t| \geq X$ and exits when the directional de-meaned gap falls to E bps or after H seconds. Each rule’s X , H and E were tuned per fee on validation, which gives the rules a per-fee advantage over the single fee-conditioned network. At every fee the convergence-exit rule beat the fixed-hold rule on validation, so it is the primary comparator; we call it the best rule (convergence exit), or the best rule for short.

6.6 Algorithms we did not use

Soft actor-critic [16] and TD3 [17] were designed for continuous action spaces; our action set has two elements, and a discrete variant of soft actor-critic would add an entropy temperature

to tune without a clear benefit. Model-based RL would have to learn $P(x' | x)$ in the kernel (7), a generative model of two order books, which is harder than the control problem. Because of exogeneity every policy can be evaluated on the same recorded paths without such a model. A learned model could still help a policy generalise beyond those paths, but it would add model error. Offline methods such as conservative Q-learning [18] penalise actions that the logging policy rarely took, to guard against distribution shift; counterfactual replay removes the policy-induced part of that shift here. Recurrent policies would address partial observability; the EMA basis covers the main hidden variable, and we kept feed-forward networks to compare the algorithms on equal footing.

7 Protocol

The protocol was written to a JSON file and registered at 02:15 UTC on 5 October 2026, before any validation result of a version 2 checkpoint or baseline. Before that, the signal analysis had used training days only, and a software check had replayed fixed rules on 60 validation windows to compare the fast replay with the simulator; that check’s P&L was not used for any choice. The protocol fixed the data splits, the fees, the candidate runs, the checkpoint selection rule, the rule grids and three success criteria. Validation choices were frozen in a selection file with its SHA-256 hash before the simulator was run on test days, and the evaluation code refuses to run if the hash does not match. Five amendments followed (Table 4; full text in Appendix C).

Table 4: Protocol amendments, with registration time (UTC, 5 October 2026) and what had been seen at that point.

No.	Time	Change	Seen before registering
1	02:30	Add a 450 ms group of three PPO runs trained, selected and tested at 450 ms, because received-book fills overstate the edge	No version 2 validation or test result
2	02:46	Add the convergence-exit rule; widen its exit grid (2b) because the best level sat on the grid edge	150 ms validation selection (PPO ahead of the fixed-hold rule by a factor of two to four)
3	02:50	Fix a floating-point error in the table’s depth VWAP (0.002% of fillable rows) and a harsher-than-simulator forced-exit price in the fast replay; rebuild tables; re-run selection, which made the same choices	150 ms validation selection; no test result
4	03:08	450 ms group primary; per-fee rank-sum checkpoint selection; paired primary endpoint; HL-clock stress; same-snapshot split; test sets labelled previously examined	Partial 150 ms test results at fees 0.5 and 1 bps (see text)
5	15:18	Add Double DQN, FQI and the bandit ablation on the 450 ms group	All PPO test results of both groups; no training of the new algorithms

Selection. The 150 ms group chose one of 54 PPO checkpoints (six runs, nine checkpoints each) by the sum of validation net P&L over the four fees. Amendment 4 changed the rule for the 450 ms group to a per-fee rank sum, so that the 0.5 bps fee, where dollar amounts are largest, does not decide alone: every checkpoint is ranked by validation net P&L at each fee, the ranks are summed, the lowest sum wins and ties go to fewer trades. Amendment 5 applied the same rule within each new algorithm. Rules were tuned per fee on validation net P&L.

Primary endpoint. For each fee, the endpoint is the per-window difference between PPO’s net P&L and the best rule’s on identical test windows, summed over windows, with a 90% interval from a bootstrap that resamples whole UTC hours of windows (10,000 resamples), so that windows close in time are kept together. A comparison is inconclusive if either policy has fewer than 100 test trades or the interval contains zero. A pooled statistic across fees was registered as supporting evidence. Amendment 5 used the same test for every algorithm against PPO and against the best rule. The original success criteria (PPO above the version 1 raw-gap rule at every fee, and trading wherever its validation P&L was positive) were kept as secondary.

Previously examined test data. The test days of 29 September to 2 October were opened by version 1’s attempt 1, which ran its threshold rule on them at fees from 0.5 to 3.5 bps. The day of 3 October was opened by version 1’s expected-value diagnostic (attempt 3 in the protocol), which made one trade. Neither set was used directly in version 2’s design, but version 1’s summary statistics and rule results on these days were known to the team, so both are previously examined, not confirmatory. Amendment 4 records that partial 150 ms test results at 0.5 and 1 bps had been seen when it made the 450 ms group primary; it does not record which policies were looked at. The simulator files for those two fees, written before the amendment, show PPO level with the convergence rule (\$107.82 against \$108.29 and \$66.07 against \$65.69; Table 9). Days from 4 October onward would be confirmatory; they were not exported in this session. A script (`latency_arb/v2/confirm.py`) evaluates the frozen PPO and rule choices of both latency groups on newly prepared days without any retraining or reselection; it does not yet cover the amendment-5 algorithms.

8 Results

8.1 Validation selection

Table 5 gives each frozen choice’s validation result at 450 ms and Figure 3 shows every candidate (Appendix D lists the Double DQN, FQI and bandit candidates). The rank sum chose PPO seed 27 at 750,000 steps; the sum of dollars would have chosen the same run at 2,000,000 steps (\$63.38 against \$62.29 summed over fees). For Double DQN it chose seed 27 at 1,000,000 steps, for FQI seed 17 and for the bandit seed 27. On validation PPO led at 0.5 and 1 bps, Double DQN and FQI were close at 0.5 bps (about \$29 each) with Double DQN ahead at 1 bps, and the bandit was far behind, the same order as on the test days. PPO’s 27 checkpoints fell in a narrow band at 0.5 bps (\$37.57 to \$41.65, with the chosen checkpoint at the top) and spread more widely at 2 and 3.5 bps, where trade counts are small.

8.2 Primary test: 450 ms, test A

Table 6 and Figure 4 give the single test run of every frozen choice in the unchanged simulator. PPO’s net P&L was positive at every fee, from \$76.91 over 5,056 trades at 0.5 bps to \$0.05 over 45 trades at 3.5 bps. A bootstrap that resamples the four test days gives 90% intervals of \$67.26 to \$86.56 at 0.5 bps, \$32.00 to \$43.75 at 1 bps, \$1.11 to \$3.31 at 2 bps and $-\$0.09$ to \$0.21 at 3.5 bps, so the 3.5 bps result cannot be told apart from zero. PPO’s net P&L was positive on all four test days at 0.5 and 1 bps, on three of four at 2 bps and on two of four at 3.5 bps. Double DQN made \$54.61, \$25.29, \$3.17 and \$0.02; FQI \$55.37, \$20.33, \$0.80 and $-\$0.06$; the bandit \$25.51, \$4.49, \$0.76 and \$0.05. The best rule made \$76.77, \$34.80, \$1.94 and \$0.05, and the version 1 raw-gap rule \$23.92, \$3.78, \$0.67 and \$0.00. The largest drawdown (peak-to-trough fall in cumulative P&L) of any learner at 0.5 or 1 bps was \$0.60 (FQI at 1 bps). The amounts are small because each trade is 0.001 BTC, about \$84; at 0.5 bps PPO’s \$76.91 is 1.8 bps of notional

Table 5: Validation (25–28 September, 1,117 windows), fill at 450 ms, frozen choices only. Net P&L in US dollars for 0.001 BTC per trade, with the number of trades. Validation numbers come from the fast replay (Section 3).

Policy	0.5 bps		1 bps		2 bps		3.5 bps	
	Net	Trades	Net	Trades	Net	Trades	Net	Trades
PPO	41.65	2,909	18.68	2,246	1.75	315	0.20	14
Double DQN	28.56	2,404	13.69	1,274	1.74	145	−0.06	4
FQI	29.14	2,440	9.78	1,653	1.12	198	0.21	7
Contextual bandit	13.22	2,924	2.48	669	0.01	41	−0.01	1
Best rule (convergence exit)	39.57	2,890	15.34	2,890	0.83	128	0.14	11
De-meaned rule, fixed hold	16.99	2,911	2.31	290	0.23	11	0.02	5
Raw-gap rule (version 1)	10.53	2,003	2.19	144	0.18	13	0.00	0

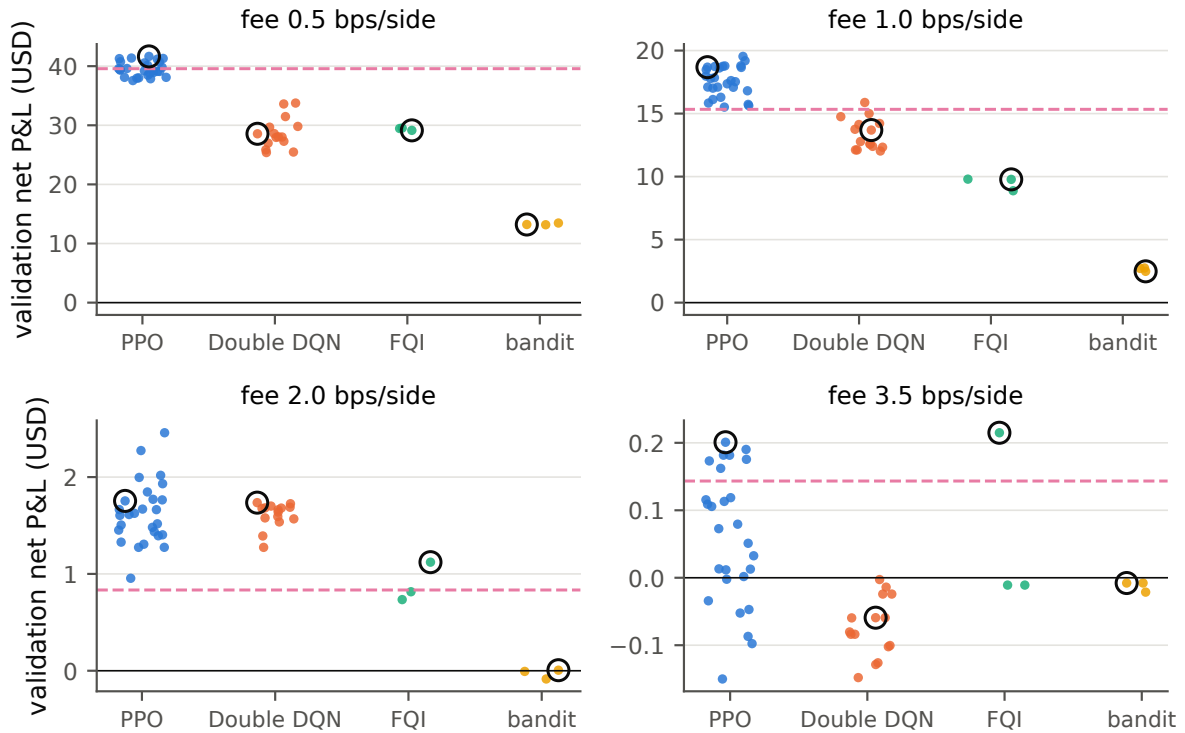


Figure 3: Validation net P&L at 450 ms of every candidate: 27 PPO checkpoints, 15 Double DQN checkpoints (five per seed), and three seeds each of FQI and the bandit. Circles mark the frozen choices; the dashed line is the best rule tuned at that fee.

per trade.

Table 6: Primary test (test A: 29 September to 2 October, 1,140 windows), fill at 450 ms, unchanged simulator. Net P&L in US dollars and number of trades. Learners above the line, rules below.

Policy	0.5 bps		1 bps		2 bps		3.5 bps	
	Net	Trades	Net	Trades	Net	Trades	Net	Trades
PPO	76.91	5,056	37.91	3,813	2.21	628	0.05	45
Double DQN	54.61	4,218	25.29	2,186	3.17	240	0.02	12
FQI	55.37	4,024	20.33	2,685	0.80	349	-0.06	27
Contextual bandit	25.51	5,081	4.49	1,193	0.76	66	0.05	5
Best rule (convergence exit)	76.77	4,972	34.80	4,972	1.94	210	0.05	30
De-meaned rule, fixed hold	28.95	4,886	4.34	508	0.61	28	0.04	11
Raw-gap rule (version 1)	23.92	2,058	3.78	158	0.67	15	0.00	0
Always flat	0.00	0	0.00	0	0.00	0	0.00	0

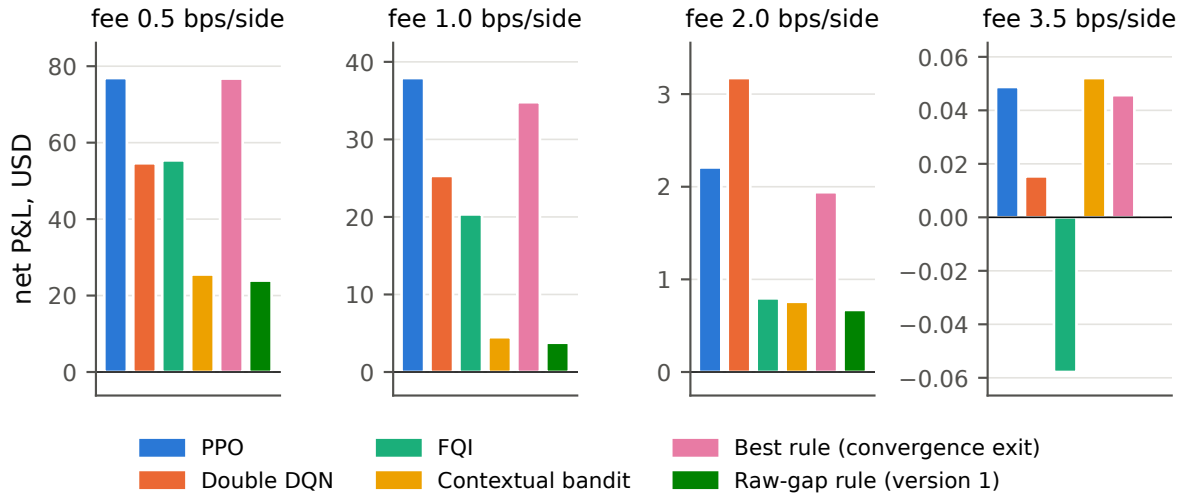


Figure 4: Primary test, net P&L by fee (each panel has its own scale). Values are in Table 6.

8.3 Paired comparisons

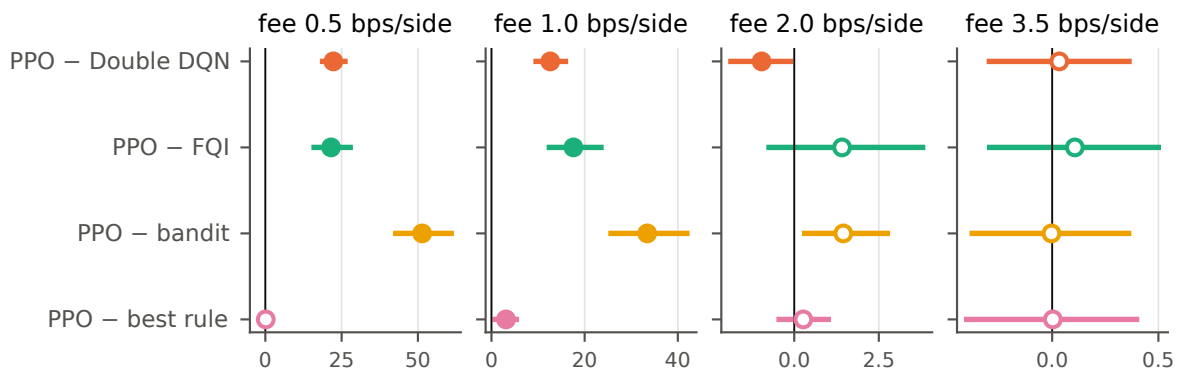
Table 7 and Figure 5 give the registered paired statistics. Against the best rule, PPO was level at 0.5 bps (\$0.14, interval $-\$1.92$ to $\$1.99$; ahead on 1 of 4 days) and conclusively ahead at 1 bps (\$3.12, interval $\$0.80$ to $\$5.34$; ahead on all 4 days and in 67 of 96 hours). At 2 and 3.5 bps the intervals contained zero. The pooled statistic sums the per-window differences over all four fees after dividing each by the rule’s validation per-window standard deviation at that fee, so one unit is one such standard deviation. It was 83.0 with an interval from -33.0 to 200.8 , so it was also inconclusive.

PPO was conclusively ahead of Double DQN, FQI and the bandit at 0.5 and 1 bps, by \$12.62 to \$51.40, and ahead on every test day in each of these six comparisons. At 2 bps Double DQN was conclusively ahead of PPO by \$0.96 (interval $\$0.12$ to $\$1.87$) and also ahead of the best rule by \$1.23 (interval $\$0.48$ to $\$2.06$), with 240 trades against PPO’s 628 and the rule’s 210. At 3.5 bps no learner made 100 trades and every comparison was inconclusive. Against the version 1 raw-gap rule PPO was ahead at 0.5 and 1 bps; at 2 and 3.5 bps that rule made 15 and 0

trades, too few for a verdict.

Table 7: Paired comparisons on the primary test (test A, 1,140 windows, 450 ms). Difference is PPO minus the other policy, summed over windows, in US dollars; the interval is the 90% hour-block bootstrap interval. “Days” counts days on which PPO was ahead. A verdict needs at least 100 trades for both policies and an interval that excludes zero. PPO’s trades were 5,056, 3,813, 628 and 45.

Fee	Other policy	Its trades	Difference	90% interval	Days	Verdict
0.5	best rule	4,972	0.14	[−1.92, 1.99]	1/4	inconclusive
	raw-gap rule (version 1)	2,058	52.98	[45.01, 61.56]	4/4	PPO better
	Double DQN	4,218	22.29	[18.76, 26.09]	4/4	PPO better
	FQI	4,024	21.54	[15.98, 27.80]	4/4	PPO better
	bandit	5,081	51.40	[42.74, 60.90]	4/4	PPO better
1.0	best rule	4,972	3.12	[0.80, 5.34]	4/4	PPO better
	raw-gap rule (version 1)	158	34.13	[26.50, 42.45]	4/4	PPO better
	Double DQN	2,186	12.62	[9.58, 15.88]	4/4	PPO better
	FQI	2,685	17.59	[12.42, 23.50]	4/4	PPO better
	bandit	1,193	33.42	[25.66, 41.95]	4/4	PPO better
2.0	best rule	210	0.27	[−0.45, 1.01]	2/4	inconclusive
	raw-gap rule (version 1)	15	1.54	[0.45, 2.72]	3/4	inconclusive
	Double DQN	240	−0.96	[−1.87, −0.12]	1/4	PPO worse
	FQI	349	1.41	[−0.75, 3.79]	2/4	inconclusive
	bandit	66	1.45	[0.31, 2.75]	3/4	inconclusive
3.5	best rule	30	0.00	[−0.40, 0.40]	2/4	inconclusive
	raw-gap rule (version 1)	0	0.05	[−0.34, 0.42]	2/4	inconclusive
	Double DQN	12	0.03	[−0.30, 0.36]	1/4	inconclusive
	FQI	27	0.11	[−0.29, 0.50]	2/4	inconclusive
	bandit	5	0.00	[−0.38, 0.36]	2/4	inconclusive



PPO minus competitor, summed over 1,140 test windows (USD); filled marker = conclusive

Figure 5: PPO minus each competitor on the primary test, with 90% hour-block bootstrap intervals. Positive values favour PPO. Hollow markers are inconclusive by the registered rule (interval contains zero, or fewer than 100 trades); the bandit at 2 bps has an interval that excludes zero but only 66 trades.

8.4 Fee-dependent selectivity

One PPO network served all four fees. Its trade count fell from 5,056 at 0.5 bps to 3,813, 628 and 45 as the fee rose (Figure 6). The best rule, tuned per fee, used the same setting at 0.5 and 1 bps (enter at 2 bps, exit at -1 bps or after 30 s) and made 4,972 trades at both fees; it became selective only at 2 bps, with 210 trades. Table 8 breaks the trades down. At 1 bps PPO made 1,159 fewer trades than the rule, earned 1.18 bps net per trade against the rule’s 0.83, and held 11.7 s on average against 13.7 s. PPO’s advantage at 1 bps came from these fewer trades at a higher net edge per trade. At 2 bps the order reversed: PPO made three times as many trades as the rule and 2.6 times as many as Double DQN, at 0.41 bps net per trade against 1.09 and 1.56. Double DQN entered less often than PPO at every fee; that cost it at 0.5 and 1 bps, where many marginal trades were still profitable, and paid off at 2 bps.

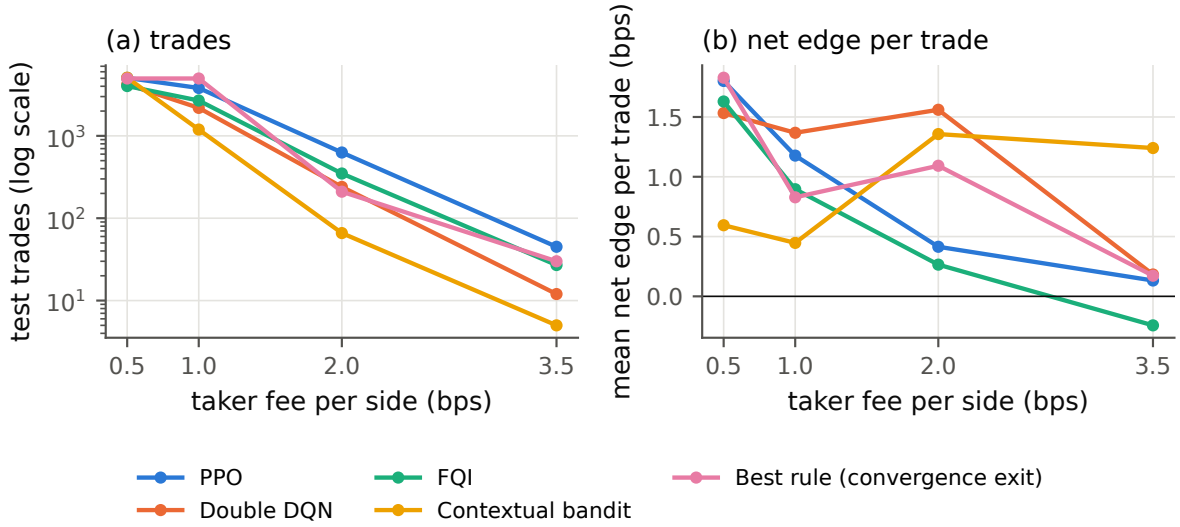


Figure 6: Primary test. (a) Trades by fee, log scale. (b) Mean net P&L per trade in bps of entry notional. At 3.5 bps the means rest on 5 to 45 trades.

Table 8: Per-trade economics on the primary test. Gross and net are mean bps of entry notional per trade; gross includes spread and slippage but not fees. “Policy exits” are exits the policy chose; “30 s exits” hit the holding limit (the rest closed at the window end). The bandit chooses no exit: every trade closes after its fixed 3 s hold.

Fee	Policy	Trades	Gross	Net	Win %	Hold (s)	Policy exit %	30 s exit %
1.0	PPO	3,813	3.18	1.18	59.9	11.7	87	10
	Double DQN	2,186	3.37	1.37	66.3	7.5	97	1
	FQI	2,685	2.90	0.90	62.6	21.5	71	24
	Bandit	1,193	2.45	0.45	55.6	3.0	fixed 3 s	
	Best rule	4,972	2.83	0.83	54.9	13.7	77	19
2.0	PPO	628	4.41	0.41	44.3	10.8	89	8
	Double DQN	240	5.56	1.56	60.0	10.9	90	7
	FQI	349	4.27	0.27	57.9	21.0	78	18
	Bandit	66	5.36	1.36	71.2	3.0	fixed 3 s	
	Best rule	210	5.09	1.09	58.1	9.1	89	10

8.5 The 150 ms group as an optimistic sensitivity

The original protocol’s 150 ms group trained six PPO runs and selected seed 27 (entropy 0.01) at its final checkpoint. With fills at 150 ms everything earned more (Table 9): PPO made \$107.82 at 0.5 bps and \$13.51 at 2 bps. Against the best rule, PPO was level at 0.5 and 1 bps and conclusively ahead at 2 bps (\$4.45, interval \$2.33 to \$6.92); at 3.5 bps the rule made only 33 trades. Amendment 4, written after an independent review and following amendment 1’s finding that received-book fills at 150 ms overstate the edge, made the 450 ms group primary and this group an optimistic sensitivity analysis. The same-snapshot split that amendment 4 registered later showed that 71% of this group’s test entries at 0.5 bps (3,797 of 5,340) filled at the HL snapshot the decision had already seen. The two groups agree that PPO and the best rule are close at 0.5 bps and disagree on whether PPO’s advantage appears at 1 or at 2 bps.

Table 9: The 150 ms group. Top: validation (selection files) and both test sets, net P&L in US dollars and trades. Bottom: paired comparison of PPO with the best rule on test A (90% hour-block bootstrap interval; same verdict rule as Table 7).

Policy	0.5 bps		1 bps		2 bps		3.5 bps	
	Net	Trades	Net	Trades	Net	Trades	Net	Trades
<i>Validation, 25–28 Sep (selection files)</i>								
PPO	60.41	3,070	35.55	2,642	5.61	498	0.57	51
Best rule (convergence exit)	58.27	2,917	33.81	2,917	4.70	302	0.49	14
De-meaned rule, fixed hold	30.50	2,964	8.41	844	1.47	132	0.33	14
Raw-gap rule (version 1)	18.61	2,008	4.01	144	1.58	144	0.14	14
<i>Test, 29 Sep–2 Oct</i>								
PPO	107.82	5,340	66.07	4,556	13.51	978	1.71	85
Best rule (convergence exit)	108.29	5,047	65.69	5,047	9.06	559	1.22	33
De-meaned rule, fixed hold	51.67	4,961	16.93	1,520	3.35	221	1.22	32
Raw-gap rule (version 1)	36.18	2,090	7.10	167	4.27	167	1.19	16
<i>Test, 3 Oct</i>								
PPO	1.44	125	0.48	106	0.13	23	0.10	3
Best rule (convergence exit)	1.39	124	0.34	124	0.15	16	0.09	3
De-meaned rule, fixed hold	1.00	123	0.35	35	0.09	5	0.04	3
Raw-gap rule (version 1)	0.65	26	0.22	3	0.17	3	0.10	3

Fee	PPO (trades)	Best rule (trades)	Difference	90% interval	Verdict
0.5	107.82 (5,340)	108.29 (5,047)	−0.47	[−2.39, 1.24]	inconclusive
1.0	66.07 (4,556)	65.69 (5,047)	0.39	[−1.97, 2.57]	inconclusive
2.0	13.51 (978)	9.06 (559)	4.45	[2.33, 6.92]	PPO better
3.5	1.71 (85)	1.22 (33)	0.49	[0.00, 1.04]	inconclusive

8.6 Test B: 3 October

On 3 October (285 windows) every policy made under 125 trades at each fee and all amounts were below one dollar (Table 10). PPO made \$0.69 at 0.5 bps and lost \$0.03 and \$0.16 at 1 and 2 bps; the best rule lost \$0.45 at 1 bps. Most paired comparisons were inconclusive because of the trade-count rule. Three were not, all at 0.5 bps: PPO was ahead of FQI (\$0.17, interval \$0.02 to \$0.34) and of the bandit (\$0.44, interval \$0.24 to \$0.69), and the best rule was ahead of

the bandit (\$0.33, interval \$0.19 to \$0.48). The day had the smallest gaps of the whole dataset (median absolute gap 0.61 bps) and says little either way.

Table 10: Test B (3 October, 285 windows), fill at 450 ms, unchanged simulator. Net P&L in US dollars and number of trades.

Policy	0.5 bps		1 bps		2 bps		3.5 bps	
	Net	Trades	Net	Trades	Net	Trades	Net	Trades
PPO	0.69	117	−0.03	83	−0.16	19	0.05	1
Double DQN	0.55	92	0.12	50	0.01	6	0.03	1
FQI	0.52	112	−0.05	64	−0.03	7	−0.02	2
Contextual bandit	0.25	109	0.01	39	0.06	3	0.03	1
Best rule (convergence exit)	0.58	121	−0.45	121	0.09	5	0.01	3
De-means rule, fixed hold	0.33	120	0.06	16	0.09	3	0.01	1
Raw-gap rule (version 1)	0.33	27	0.15	3	0.10	3	0.00	0

8.7 Learning curves

Figure 7 shows training on the 450 ms group. In its first rollout (4,096 steps, before smoothing) PPO took about half of the screened entries and lost 8.6 to 9.7 bps of notional per window; the smoothed curves in the figure start at less extreme values because each point averages up to ten log points. Its entry share then fell to between 2% and 5% (smoothed) after 61,000 to 78,000 steps, the same move toward abstention that version 1 made, but here it recovered: training reward per window turned positive after 61,000 to 82,000 steps, and after one million steps the entry share settled at 21% to 24% and the reward at 2.0 to 2.1 bps per window for all three seeds. Unlike version 1, the runs logged PPO’s optimisation statistics (Appendix H): after one million steps the approximate Kullback–Leibler (KL) divergence between successive policies averaged about 0.001 per update and the value function explained about 56% of the return variance. Double DQN’s entry share fell from about one half to 17% to 18% as ϵ decayed, its reward turned positive after 180,000 to 190,000 steps and settled at 1.4 to 1.5 bps per window, still with 2% random actions. The three seeds of each algorithm followed nearly the same path. Training rewards are not comparable across the two algorithms, because PPO samples from a stochastic policy and DQN acts ϵ -greedily.

Figure 8 shows how the test P&L accumulated. At 1 bps PPO and the best rule rose in steps during the same busy periods, and PPO pulled ahead at the larger steps. At 2 bps most of the differences arose on 30 September, when Double DQN made \$1.53 and FQI lost \$0.35 (Appendix F).

9 Stress tests

The frozen 450 ms choices were replayed under harsher assumptions without reselection (Table 11, Figure 9; full grid in Appendix G).

Latency. Every additional 150 ms of fill delay cost PPO about \$15 to \$18 at 0.5 bps. At 600 ms PPO still made \$58.96 and \$23.88 at 0.5 and 1 bps, but lost \$0.61 at 2 bps, where the best rule kept \$0.91.

Spread. With every book level’s distance from the mid scaled by 1.5, PPO made \$71.11, \$33.42, \$1.21 and −\$0.05.

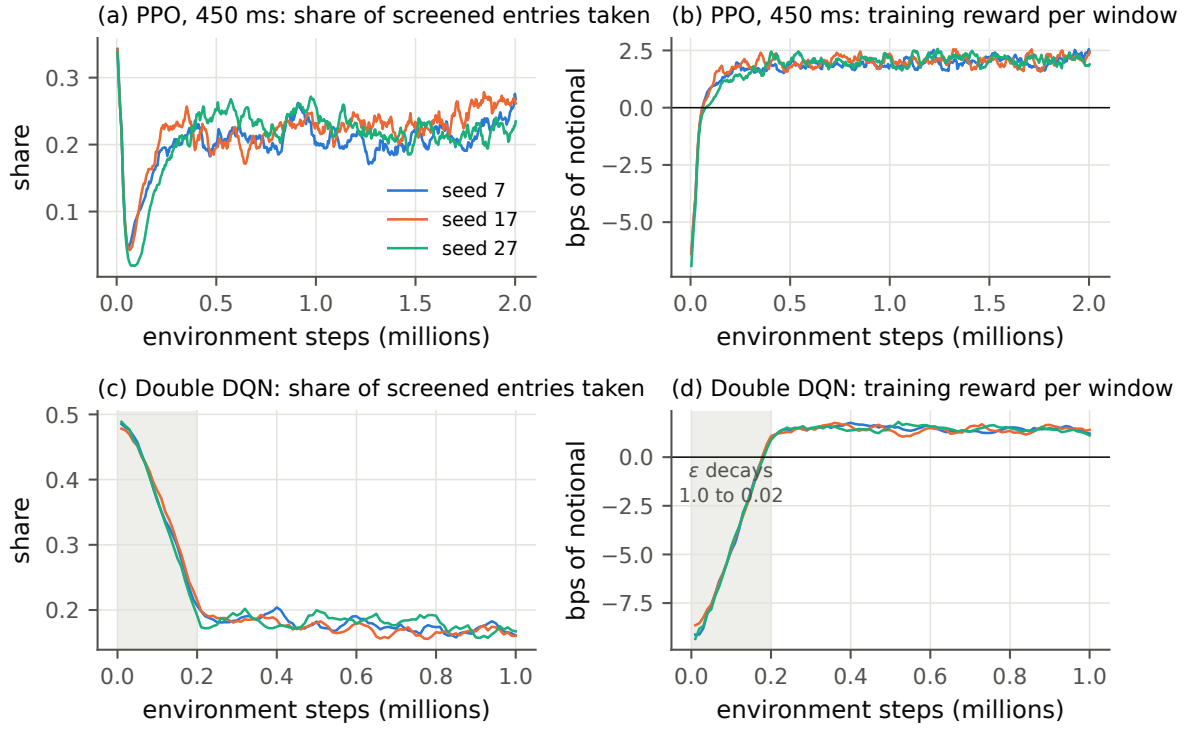


Figure 7: Training curves, 450 ms fills, three seeds each, smoothed over ten log points (five for DQN). Entry share is the fraction of screened flat decisions on which the agent entered. The shaded band in (c) and (d) is the ϵ -decay phase. The 150 ms PPO curves are in Appendix H.

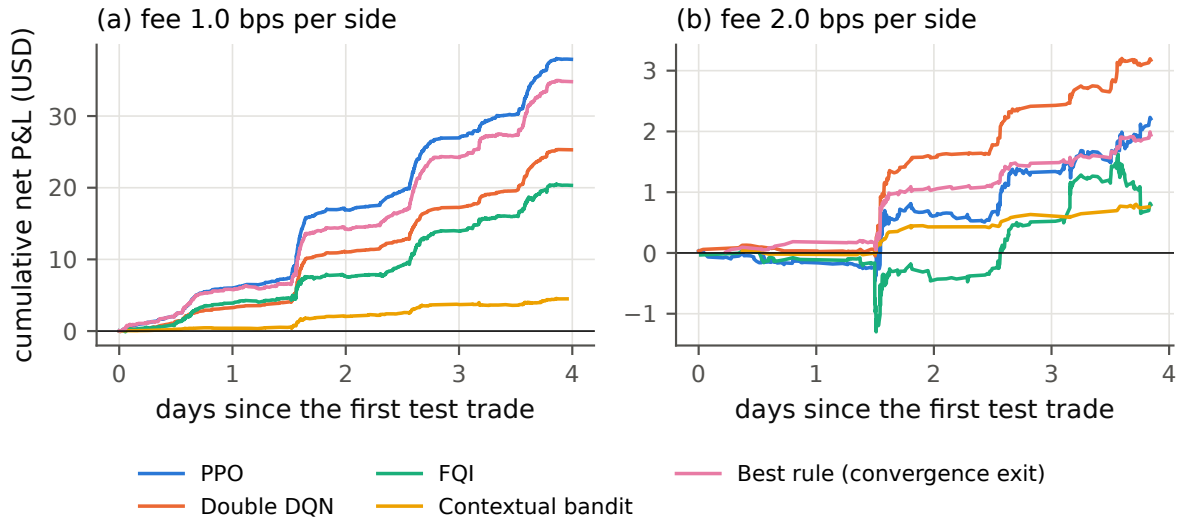


Figure 8: Cumulative net P&L over the primary test, trades ordered by exit time. Per-day values are in Appendix F.

Fee 4.5 bps (extrapolation). PPO was given a fee of 4.5 bps, outside its training range, and the rules kept their 3.5 bps settings. PPO cut its trades to 13 and lost \$0.10; the best rule made 30 trades and lost \$0.46; the raw-gap rule did not trade.

HL-clock fills. This is the strictest convention we could compute (Table 12); at 450 ms it combines the order delay with HL’s publication lag (Section 3.2). At 0.5 bps PPO made \$27.05, the best rule \$26.11, FQI \$17.86, Double DQN \$11.58 and the raw-gap rule \$4.23, while the bandit lost \$13.69 and the fixed-hold rule \$4.09. At 1 bps and above every policy that traded lost money, including PPO (−\$4.62, −\$10.07 and −\$2.21). On 3 October every policy that traded lost money at every fee under this convention, except the raw-gap rule’s three trades at 1 bps (\$0.01).

Same-snapshot share. Table 13 splits PPO’s received-book results by whether the entry filled at the same HL snapshot the decision saw. At 450 ms 19% of entries did, and they produced \$19.27 of the \$76.91 at 0.5 bps. At 150 ms the share was 71%, at 600 ms 1%. Under HL-clock fills it was 1% (45 of 5,049 entries).

Table 11: Stress tests on test A for the frozen 450 ms choices: PPO / best rule, net P&L in US dollars (fast replay; its 450 ms column equals the simulator’s). Trade counts are in Appendix G.

Fee	150 ms	300 ms	450 ms	600 ms	Spread $\times 1.5$	HL clock
0.5	107.51 / 108.29	91.55 / 91.53	76.91 / 76.77	58.96 / 58.24	71.11 / 70.86	27.05 / 26.11
1.0	63.36 / 65.69	50.27 / 49.22	37.91 / 34.80	23.88 / 16.32	33.42 / 28.89	−4.62 / −15.88
2.0	10.34 / 6.13	7.09 / 4.34	2.21 / 1.94	−0.61 / 0.91	1.21 / 1.48	−10.07 / −3.74
3.5	1.50 / 1.22	1.14 / 0.92	0.05 / 0.05	0.19 / 0.06	−0.05 / −0.03	−2.21 / −1.37

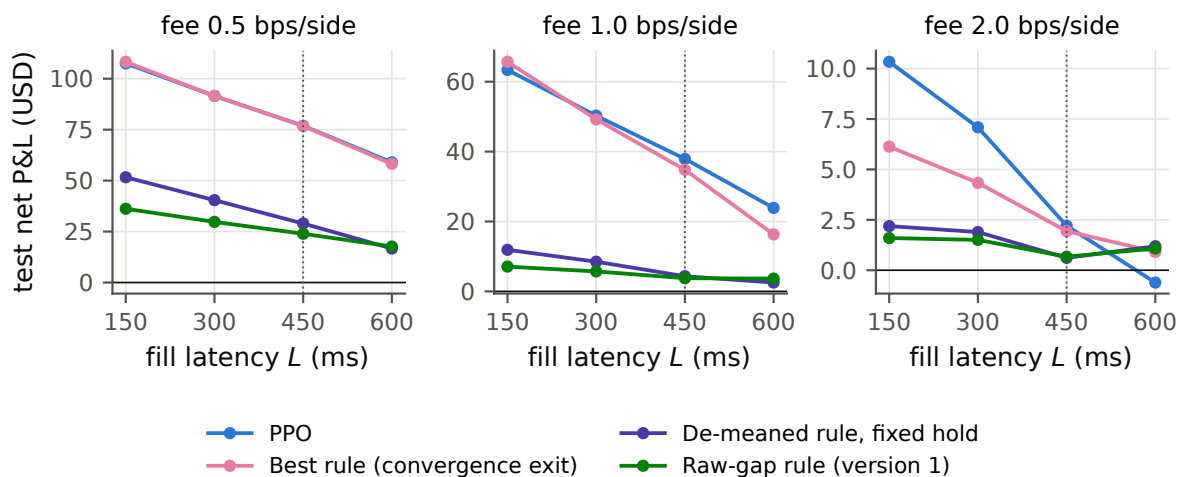


Figure 9: Test net P&L of the frozen 450 ms PPO choice and the rules when the same decisions are filled 150 to 600 ms after the decision (received-book convention). The dotted line marks the primary latency.

10 Discussion: what the algorithms learned

10.1 A path-dependent exit was worth more than a fixed hold

The bandit and FQI share the entry method and differ in the exit. With its fixed 3 s exit the bandit made \$25.51 and \$4.49 at 0.5 and 1 bps, against FQI’s \$55.37 and \$20.33. Their entry

Table 12: HL-clock fills at 450 ms on test A: fills priced from the last HL book whose exchange timestamp is at or before $t + 450$ ms. Net P&L in US dollars and trades.

Fee	PPO		Double DQN		FQI		Bandit		Best rule		Raw-gap rule	
	Net	Tr.	Net	Tr.	Net	Tr.	Net	Tr.	Net	Tr.	Net	Tr.
0.5	27.05	5,049	11.58	4,233	17.86	3,850	-13.69	5,082	26.11	4,974	4.23	2,056
1.0	-4.62	3,806	-4.50	2,179	-6.92	2,581	-10.44	1,184	-15.88	4,974	-0.27	157
2.0	-10.07	621	-2.88	237	-5.06	334	-1.41	65	-3.74	212	-0.18	15
3.5	-2.21	44	-0.60	12	-1.03	27	-0.25	5	-1.37	29	0.00	0

Table 13: PPO entries (450 ms choice, test A) that filled at the same HL snapshot the decision saw, by fill latency, with the net P&L those trades produced out of the total (received-book convention).

Latency (ms)	Fee 0.5 bps		Fee 1 bps	
	Same-snapshot entries	Their net (\$)	Same-snapshot entries	Their net (\$)
150	3,644 of 5,117 (71%)	83.36 of 107.51	2,915 of 3,894 (75%)	51.39 of 63.36
300	2,073 of 5,085 (41%)	44.64 of 91.55	1,721 of 3,853 (45%)	27.52 of 50.27
450	967 of 5,056 (19%)	19.27 of 76.91	813 of 3,813 (21%)	11.37 of 37.91
600	71 of 5,032 (1%)	1.43 of 58.96	65 of 3,783 (2%)	0.92 of 23.88

regressions were fitted to different payoffs, so they also entered on different seconds (1,193 against 2,685 trades at 1 bps), and this comparison does not separate the exit from the entry. On the 79,842 training paths, the same paths the exit network was fitted to, FQI’s learned exit raised the mean payoff over all admissible entries from -2.67 to -2.69 bps under the 3 s exit to -1.63 to -1.68 bps (ranges over seeds), about 1 bps per trade before any entry selection; this is an in-sample figure. Among the rules, both de-meaned rules entered at the same 2 bps threshold at 0.5 bps and made similar numbers of trades (4,972 and 4,886). There the convergence exit made \$76.77 against the fixed hold’s \$28.95, 2.7 times as much. At 1 bps the tuned rules differ in threshold as well as exit (2 against 4 bps; 4,972 against 508 trades), so their ratio (\$34.80 against \$4.34) does not measure the exit alone. A path-dependent exit, whether learned or hand-written, was worth much more than a fixed hold; we did not isolate the contribution of learning the exit.

10.2 PPO against a hand-written convergence rule

At 0.5 bps PPO and the best rule were indistinguishable: similar trade counts (5,056 and 4,972), similar net edge per trade (1.80 and 1.83 bps) and a paired difference of 14 cents over four days. At that fee a three-parameter rule tuned on four validation days captured what PPO learned from 22 training days. PPO’s advantage appeared at 1 bps, where it entered more selectively than a rule whose validation tuning kept the 0.5 bps setting: the fee-conditioned network cut its trades by 25% between 0.5 and 1 bps, while the rule’s grid search returned the same setting at both fees. The difference is \$3.12 over four days.

10.3 Double DQN’s selectivity at 2 bps

Double DQN entered less often than PPO at every fee. At 0.5, 1 and 3.5 bps it also held for shorter times (4.8 s against 10.8 s, 7.5 against 11.7 s and 10.2 against 12.3 s); at 2 bps the holds were about equal (10.9 against 10.8 s). Its win at 2 bps, where most screened entries no longer cover the fee, came from fewer entries with a higher edge: 240 trades at 5.56 bps gross against

PPO’s 628 at 4.41 bps. Low-fee windows carry most of PPO’s training reward and dominate its gradient, which may explain its extra trades at 2 bps. The Double target may explain DQN’s caution, because it reduces the upward bias in the targets (Section 6.2). We did not run plain DQN, so we cannot attribute the difference to the target. Only the selected Double DQN, FQI and bandit policies were run on the test days, so the spread between seeds and checkpoints on test is known for PPO only (Appendix E); we cannot say whether the 2 bps result holds for other checkpoints of either learner.

10.4 What FQI’s greedy entry and fitted exit give up

FQI trained in under two minutes per seed and reached 72% of PPO’s test P&L at 0.5 bps and 54% at 1 bps. Two simplifications cost it. First, its entry rule enters whenever the predicted payoff is positive and so ignores the option value of waiting and the cost of being busy (Equation (14)). Second, its exit held too long: 21.5 s on average at 1 bps against PPO’s 11.7 s, with 24% of trades reaching the 30 s limit against 10%. On the training paths the fitted value at the first holding second averaged 0.83 to 0.98 bps (computed from the network before its last fit), while the same paths under the learned exit realised -1.63 to -1.68 bps. Four causes are consistent with this gap of about 2.5 bps and with the long holds. The targets in Equation (17) take a maximum over fitted values, which is biased upwards and inflates the value of holding; Longstaff–Schwartz targets avoid this bias and the Double DQN target reduces it. The stopping problem leaves out the value of being flat after an exit (Section 4.7), which lowers the value of exiting. The Huber loss fits a robust central value, which differs from the mean when payoffs are skewed. And 15 iterations of one pass each are fewer than the 29 backward steps that exact backward induction over the holding horizon would need. We did not run the experiments that would separate these causes.

10.5 When supervised or bandit methods would be enough

If the exit is fixed, the entry problem is close to a contextual bandit (Section 4.7), and a regression on counterfactual payoffs is the natural method. The bandit ablation did poorly with its fixed 3 s exit. A bandit with the convergence exit, the version the protocol registered, was not run and might close much of the gap. The evidence here suggests that a path-dependent exit pays (Section 10.1) and that a learned, fee-conditioned policy helps when one policy must serve many cost levels (the 1 bps result). For a single known fee, a tuned rule with a convergence exit matched PPO.

11 Would faster order-book feeds help?

If the HL-clock convention is right, the estimates below say yes, mainly through two changes: deciding on Binance events instead of whole seconds, and a faster view of HL’s book. A faster Binance stream alone adds little. The numbers come from the sub-second analysis on six training days (4, 8, 12, 15, 19 and 23 September) with a fixed 3 s exit, so they estimate what a faster feed could add to a fixed 3 s rule; no learned agent was tested on sub-second data.

The edge is perishable. At a 3 bps threshold the gross edge fell from 3.86 bps with an immediate fill to 3.41 at 150 ms, 2.98 at 300 ms, 2.48 at 500 ms and 1.02 at 1,000 ms, about 0.28 bps per 100 ms of delay (Figure 10).

The one-second grid misses and delays dislocations. A dislocation, the time the de-meaned gap stays beyond the threshold, lasted 618, 510 and 306 ms at the median for thresholds of 3, 5 and 8 bps. The grid never saw 28.5%, 36.0% and 47.6% of these onsets within 10 s, and

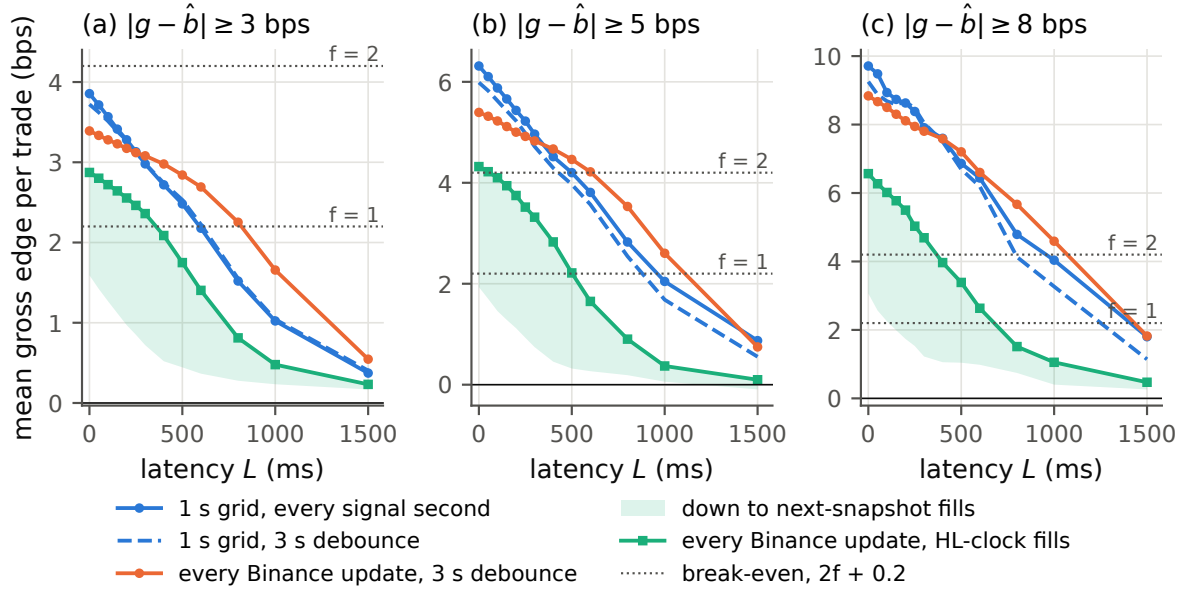


Figure 10: Sub-second analysis on six training days: mean gross edge of a 3 s round trip on HL against execution latency, for de-meaned gaps of at least 3, 5 and 8 bps. Blue: one-second decision grid. Orange: decisions at every Binance update. Green: decisions at every Binance update with HL-clock fills; the shaded band extends down to next-snapshot fills. Solid blue counts every signal second; dashed blue and the event-driven curves apply a 3 s debounce (no new signal is counted within 3 s of the previous one). Received-book fills except where marked. In this analysis gross edge excludes slippage, so the dotted break-even lines sit at twice the fee plus 0.2 bps, for fees f of 1 and 2 bps per side. Redrawn from `analysis/subsecond_edge.json`.

when it did see one it added 452 ms of delay at the median. Under received-book fills this delay looks almost free (0.06 bps per trade), because the received book is itself stale. Under HL-clock fills, waiting for the grid cost 1.0 to 1.9 bps per trade. Daily net P&L summed over trades at 1 bps per side and a 3 bps threshold, with HL-clock fills, was -78 bps for the grid, $+330$ bps for event-driven decisions and $+538$ bps for event-driven decisions with an emulated zero-delay view of HL’s book.

Event-driven decisions change the epochs, the discount and the state’s clock. In the model of Section 4, deciding on every Binance update replaces the whole-second epochs with Binance event times: the screen (4) is evaluated at each update, sojourn times become random intervals of about 100 ms for the depth stream, and the per-epoch discount in the objective (10) should become γ^{τ_k} . The market features are built on a one-second clock (the EMA’s per-second gain, 1 s and 5 s returns, 30 s volatility, holding time in seconds) and would have to be redefined on an event clock, and holding epochs would become about ten times as many. The action sets, the factored transition (5) and the reward (8) keep their form. Training cost grows with the number of epochs; the screen limits flat epochs to updates during dislocations.

Binance bookTicker and HL’s best-bid-offer stream. At a 3 bps threshold with HL-clock fills, the grid earned 2.04 bps per trade against an ideal of 3.36, losing 1.32 bps. Deciding on every Binance depth update would recover about 0.60 bps of that (45%). Binance’s bookTicker stream, if it showed moves 50 ms sooner than the 100 ms depth stream, would add about 0.08 bps (6%), at most 0.16 bps (12%). A faster view of HL’s book would add about 0.45 bps (34%) and would also remove the 19% of signals that HL had already closed. An informal 15 s probe by the team on 5 October, outside the dataset, counted about 446 messages per second on Binance

btcdsdt@bookTicker and about 5.9 per second on HL’s best-bid-offer stream, against the 1.84 12Book snapshots per second we recorded. A faster HL top-of-book stream is the more promising of the two, but we have not recorded it, and its timing relative to HL’s matching engine is unknown.

What colocation and kernel bypass can and cannot fix. Colocation, placing servers in the same data centre or region as the venue, shortens the network part of the order path and of our receive delay, which lowers L . Kernel-bypass networking, which moves packets without the operating system’s network stack, saves microseconds, while the delays that matter here are hundreds of milliseconds. Neither changes how often HL publishes its book (every 540 ms at the median) or the lag between HL’s timestamp and the book reaching subscribers, and neither changes the fee. Faster infrastructure also invites the competition that Budish et al. [1] describe: other fast traders react to the same Binance move, and the HL quote that our replay treats as available may already be gone.

12 Responsible AI

12.1 Fairness and market effects

Latency arbitrage earns money by trading against quotes that have not caught up with the market. The other side is usually a liquidity provider whose quote is stale. Budish et al. [1] argue that this race is a cost to liquidity providers that is passed on to all traders through wider spreads. A working version of our agent would be on the fast side of that transfer. The amounts we found for a small taker are modest and depend on low fees, but the concern applies to anyone with lower fees or faster infrastructure. Nothing in this project connects to a live market or uses capital.

12.2 Transparency

The protocol and every amendment are dated files; the amendments state what had been seen when they were written. Selection files are hash-frozen before testing. Every table and figure in this report is produced from saved run files by a script stored with the report, and Appendix A lists those files. Failed experiments and deviations from the protocol are reported where they occur (Sections 5, 6 and 7).

12.3 Accountability

Policies were selected on validation days only and tested once. Risk limits (position size, holding limit, freshness checks) are enforced by the environment, not left to the learned policy. The AI-use statement records which parts AI agents produced; the team is responsible for reviewing them.

12.4 Trust and calibrated claims

The main risk of misuse is over-reading the results; Section 13 lists the assumptions they rest on. To limit over-reading, we report trade counts next to every P&L figure, report every run and stress result whatever the outcome, and use a registered rule that declares comparisons with few trades inconclusive.

12.5 Use of AI coding agents

AI agents wrote much of the code that produced these numbers. Their errors would be silent: a look-ahead in a feature or a mismatch between the fast replay and the simulator would inflate

results without any visible failure. The project’s safeguards were unit tests that compare the fast replay with the simulator trade for trade; an independent review of the code and protocol (three reviewers, with each finding checked by a separate reviewer), which found no look-ahead and led to amendments 3 and 4; and reporting the primary test from the unchanged simulator. These reduce but do not remove the risk, and the team should review the code before relying on it.

13 Limitations and threats to validity

Reused test data. Both test sets were opened by version 1, and partial 150 ms test results were seen before amendment 4 made the 450 ms group primary. The algorithm comparison was registered after all PPO test results were known. No result here is confirmatory; days from 4 October onward are needed.

Fill convention. All headline numbers use received-book fills, which we showed to be optimistic. HL-clock fills assume that our clock matches HL’s and that HL’s timestamp marks when the book was valid; neither was verified. At 450 ms they also assume a 450 ms order path on HL’s clock, longer than the 150 ms behind the primary choice (Section 3.2). Under HL-clock fills the edge survived only near 0.5 bps per side.

Market model. Orders take displayed liquidity without moving later prices, other traders are not modelled, the fill delay is fixed, and funding is set to zero. The position is 0.001 BTC; larger orders would walk deeper into HL’s five displayed levels.

Conditioning on clean windows. Windows are kept only if all five minutes pass quality checks, so every policy implicitly knows that no feed outage occurs later in its window.

Unequal tuning. Rules were tuned per fee on four validation days, while the learners were fitted on 22 training days and selected on validation. Each learner used one hyperparameter configuration (PPO had two entropy settings at 150 ms) and three seeds; deep RL results can depend strongly on both [19].

Implementation asymmetries. PPO was trained on the table built before the amendment-3 fix (389 of about 14.3 million training fill cells differed); Double DQN, FQI and the bandit used the rebuilt table. Discounting is per epoch rather than per second. The bandit’s exit deviates from its registration, and FQI ran fewer iterations than the holding horizon.

Scope. One pair of venues, one month, one instrument whose exact contract identity on Binance (spot or futures endpoint) the recorded rows do not state. The market changed between the splits (Table 1), and the cause of the 18–23 September basis was not established.

14 Conclusion and next steps

The evidence suggests that version 1’s PPO agents abstained because the raw gap carried no usable edge on the training days: it mixed a slow basis with the lead-lag term, and entering on it with a 3 s exit lost money before fees at every threshold from 1 to 10 bps. Writing the task as an SMDP made the fix and the algorithm choices explicit. A causal EMA of the basis acts as an approximate belief state; exogenous market dynamics give counterfactual evaluation without policy-induced distribution shift, an optimal-stopping exit and a near-bandit entry. With these changes PPO traded. On reused test days with 450 ms fills its net P&L after modelled costs was positive at 0.5, 1 and 2 bps per side and could not be told apart from zero at 3.5 bps. It was conclusively ahead of Double DQN, FQI and a contextual bandit at 0.5 and 1 bps, level with the best hand-written rule at 0.5 bps and ahead of it at 1 bps; Double DQN was ahead of the selected

PPO policy at 2 bps. A path-dependent exit, learned or hand-written, was worth much more than a fixed hold; we did not isolate the contribution of learning it. Under HL-clock fills the edge survived only near 0.5 bps per side, so the result does not show a deployable strategy.

Next steps, in order of what they would settle:

- Export days from 4 October onward and run `confirm.py` on the frozen choices, after extending it to Double DQN and FQI. This is the only way to obtain confirmatory results.
- Train and select under HL-clock fills, so that the agent learns which signals survive the stricter convention instead of being stress-tested on it.
- Move decision epochs to Binance events and record HL's best-bid-offer stream, which the sub-second analysis identifies as the largest recoverable sources of edge.
- Run the registered bandit (convergence exit) and a plain DQN. Fit FQI's exit with Longstaff–Schwartz targets (the realised payoff under the current stopping rule) by exact backward induction over $h = 29, \dots, 1$, and include the value of being flat after an exit.
- Record funding rates and measure the order path latency to HL instead of assuming it.
- Export the primary 450 ms policy, as was done for the 150 ms policy (Appendix J), before any live test.

Team roles

Roles to be confirmed by each member. The proposal planned the following division; the AI-use statement below records which parts AI agents implemented.

- *Kerk Tai Heng*: data preparation and episode generation.
- *Member 2*: replay environment and transaction-cost model.
- *Member 3*: baseline strategy and evaluation tools.
- *Member 4*: PPO training and tuning.
- *Member 5*: held-out evaluation and fee sensitivity tests.
- *Member 6*: Responsible-AI analysis, report integration and presentation.

TODO (team): Replace the placeholders and edit each line so that it states what each member actually did in versions 1 and 2, including the algorithm comparison. Changes from the proposal are allowed but should be stated.

AI-use disclosure

AI coding agents did a large share of the engineering and writing in this project, under the team's direction. An OpenAI Codex-based coding agent and Anthropic's Claude implemented substantial parts of the pipeline across both versions: data preparation and quality checks, the replay environment and cost model, the version 2 decision table and fast replay, the PPO, Double DQN and fitted-Q training code, the rules, the experiment drivers, the selection and paired-statistics code, the sub-second analysis and the run reports. Claude also reviewed saved run artifacts against the source code, wrote the figure and table script for this report (`make_report_figures.py`) and drafted this report's text. The research question, the data recording, the cost assumptions and the evaluation plan come from the team's proposal, and the redesign decisions were made under the team's direction. The team reviewed the agents' output and is responsible for the code, the results and this report. ChatGPT was used to edit the proposal for clarity.

TODO (team): Confirm or edit this statement so that it matches exactly what each member and each AI tool did in versions 1 and 2, including which parts of the code and text the team wrote or rewrote, before submission.

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Appendices

A Reproducibility, provenance and compute

All runs used Python 3.12.3, NumPy 2.5.3, PyTorch 2.14.1 (CPU), Gymnasium 1.3.0 and Stable-Baselines3 2.9.0. Saved artifacts:

- Data: `data/v2-recorded-30d/` and `data/v2-fresh-2026-10-03/`, prepared by `runs/v2_prepare_data.py` with version 1's unchanged preparation code; per-second tables in `data/v2-tables/`.
- 150 ms PPO group: `runs/v2-leadlag-ppo/` (protocol, amendments 1–5, selection files with hashes, simulator results in `sim/`, fast-replay results in `fast/`, paired statistics in `stats.json`).
- 450 ms PPO group (primary): `runs/v2-leadlag-ppo-lat450/`, same layout.
- Amendment-5 algorithms: `runs/v2-algos-lat450/`: Double DQN runs in `dqn/`, FQI and bandit in `batch/`, the hash-frozen `selection.json`, simulator results in `sim/`, HL-clock results in `fast/` and paired statistics in `stats.json`.
- Analyses: `docs/report_v2/analysis/` (training-day signal diagnosis and the sub-second study, each with its script, JSON output and figures).

The bodies of the results tables (Tables 5 to 13 and the appendix result tables) and every figure except Figures 10 and 13 are written by `docs/report_v2/make_report_figures.py`, which reads only the files above and also writes `generated/report_numbers.json` with the numbers quoted in the text. Figures 10 and 13 are report-sized redraws by `docs/report_v2/make_subsecond_figures.py` of the sub-second analysis figures; the first uses only the analysis's JSON output, and the second recomputes its histograms read-only from the recorded data and checks them against that output. The data, settings, algorithm and protocol tables were written by hand from the protocol files, the run settings and version 1's report. Wall-clock training time per run: 1,217 to 1,229 s for the six 150 ms PPO runs, 1,591 to 1,613 s for the three 450 ms PPO runs, 1,171 to 1,176 s for the three Double DQN runs and 95 to 104 s for each FQI and bandit seed; about 4.4 hours in total on CPU.

Two documentation differences. Amendment 4 named the rank-sum selection file `selection_primary.json`, but the 450 ms group's `selection.json` records both the rank-sum choice (used) and the sum-of-dollars choice. The version 2 design note quotes different training-day signal numbers (for example 3.30 bps gross at a 3 bps threshold, against 2.91 bps in the stored analysis output); we could not trace the difference and use the stored output throughout.

B Hyperparameters

Table 14: Training settings. All learners share the 20-feature observation with fixed scales, the screen (de-measured gap of at least 2 bps, 30 s of history, fresh HL quote), the training fees 0.5 to 3.5 bps sampled per window, and the 22 training days.

Learner	Settings
PPO	Stable-Baselines3 2.9; 8 environments $\times$ 512 steps per update; minibatch 512; 10 epochs; learning rate 3×10^{-4} ; $\gamma = 0.99$; GAE $\lambda = 0.95$; clip 0.2; value coefficient 0.5; gradient-norm clip 0.5; advantage normalisation; entropy coefficient 0.01 (and 0.03 for three 150 ms runs); separate 64×64 tanh policy and value networks; 2,002,944 steps; checkpoints every 250,000 steps; seeds 7, 17, 27.
Double DQN	Stable-Baselines3 DQN with the Double target (16); Huber loss; 64×64 ReLU Q-network; 1,000,000 steps; buffer 200,000; minibatch 256; learning rate 10^{-4} ; 10,000 warm-up steps; one gradient step per 4 environment steps; hard target update every 5,000 steps; ϵ from 1.0 to 0.02 over the first 200,000 steps; gradient-norm clip 10; $\gamma = 0.99$; checkpoints every 250,000 steps; seeds 7, 17, 27.
FQI exit	79,842 paths (39,921 admissible training entries, each with two fees drawn from the training set); two-headed $20 \rightarrow 64 \rightarrow 64 \rightarrow 2$ tanh network; Huber loss ($\beta = 2$); Adam, learning rate 10^{-3} ; minibatch 4,096; 15 iterations of one epoch each, network kept between iterations; undiscounted; payoffs in bps of entry notional; seeds 7, 17, 27.
FQI entry and bandit	$20 \rightarrow 64 \rightarrow 64 \rightarrow 1$ tanh regression of the realised payoff (learned exit for FQI, fixed 3 s exit for the bandit) on the flat observation; 8 epochs; enter when the prediction is positive.

C Protocol history in full

Table 15: Registered files in runs/v2-leadlag-ppo/ (copies in the other run directories). Times are UTC on 5 October 2026.

File	Time	Content
protocol.json	02:15	150 ms group: six PPO runs (seeds 7, 17, 27 $\times$ entropy 0.01, 0.03), checkpoints every 250,000 steps; selection by the sum of validation net P&L over the four fees, ties to fewer trades; raw-gap rule ($X \in \{1, 2, 4, 7, 10, 20\}$) and fixed-hold de-meaned rule ($X \in \{2, \dots, 20\}$, $H \in \{1, 2, 3, 5, 10, 30\}$) tuned per fee; stress at 300, 450, 600 ms, spread $\times 1.5$ and fee 4.5 bps; success criteria S1 (PPO above the raw-gap rule at every fee), S2 (PPO trades wherever its validation P&L is positive), S3 (ablation against the de-meaned rule, reported whatever the outcome); all runs and stress results reported.
Amendment 1	02:30	450 ms group (seeds 7, 17, 27, entropy 0.01) trained, selected and tested at 450 ms, motivated by the sub-second finding that received-book fills overstate the edge (3.39 against 2.04 bps at a 3 bps threshold).
Amendment 2	02:46	Convergence-exit rule ($X, E \in \{-1, 0, 0.5, 1\}$, $H \in \{3, 10, 30\}$), added after the 150 ms selection showed PPO ahead of the fixed-hold rule by a factor of two to four on validation. 2b (02:47): E grid widened to $\{-4, -3, -2, -1.5, -1, -0.5, 0, 0.5, 1\}$ because the best E was -1 , the grid edge, at every fee; first grid kept in selection_extra_grid1.json.
Amendment 3	02:50	Review fixes: floating-point cancellation in the table’s depth VWAP (0.002% of fillable rows marked too thin) and a forced-exit price that charged the 25 bps penalty on the whole quantity in thin books. Both affected only the fast replay. Tables rebuilt; selection re-run with the same choices; earlier selections kept as *_table_v1.json; a guard refuses simulator evaluation at any cadence other than one second.
Amendment 4	03:08	From an independent adversarial review (14 findings, 10 major): 450 ms group primary; per-fee rank-sum selection; paired primary endpoint with hour-block bootstrap, per-day table and sign tests; inconclusive below 100 trades or when the interval contains zero; pooled statistic in validation standard-deviation units; HL-clock fill stress; same-snapshot split; fee 4.5 bps labelled extrapolation; test sets labelled previously examined; two limitations added (clean-window conditioning, unequal tuning). Registered after partial 150 ms test results at 0.5 and 1 bps had been seen.
Amendment 5	15:18	Double DQN, FQI with greedy entry, and the contextual-bandit ablation on the 450 ms group, with the settings in Appendix B; same selection rule; same paired tests against PPO and the best rule; all results reported. Registered after all PPO test results, before any training of these algorithms.

Selection files were frozen at 02:59:48 UTC (150 ms group, rules), 03:00:23 UTC (150 ms convergence rule), 04:42:26 UTC (450 ms group) and 04:42:50 UTC (450 ms convergence rule). Against the original success criteria at 450 ms: S1 held in net terms at all four fees (\$76.91, \$37.91, \$2.21, \$0.05 against \$23.92, \$3.78, \$0.67, \$0.00), with conclusive paired differences only at 0.5 and 1 bps; S2 held (PPO traded at every fee and its validation P&L was positive at every fee).

D Validation results of every algorithm candidate

Table 16: Validation (450 ms) of every Double DQN checkpoint and every FQI and bandit seed: net P&L in US dollars and trades. * marks the frozen choice. “final” is the model saved at the end of training.

Learner	Seed	Steps	0.5 bps		1 bps		2 bps		3.5 bps	
			Net	Tr.	Net	Tr.	Net	Tr.	Net	Tr.
Double DQN	7	250,000	33.76	2,817	15.00	1,624	1.58	170	−0.08	9
Double DQN	7	500,000	26.97	2,517	14.22	1,482	1.70	150	−0.13	7
Double DQN	7	750,000	28.06	2,576	14.75	1,537	1.72	158	−0.13	10
Double DQN	7	1,000,000	25.47	2,312	12.59	1,213	1.66	145	−0.08	11
Double DQN	7	final	25.39	2,302	12.57	1,206	1.68	144	−0.08	11
Double DQN	17	250,000	33.61	2,882	15.88	1,700	1.27	225	−0.15	25
Double DQN	17	500,000	28.02	2,607	12.33	1,375	1.39	198	−0.06	13
Double DQN	17	750,000	27.30	2,309	12.03	1,071	1.57	137	−0.01	3
Double DQN	17	1,000,000	29.81	2,587	12.12	1,151	1.64	141	−0.02	10
Double DQN	17	final	29.68	2,575	12.11	1,138	1.68	141	−0.02	10
Double DQN	27	250,000	31.47	2,873	14.13	1,519	1.68	151	−0.10	7
Double DQN	27	500,000	27.95	2,775	12.78	1,572	1.54	162	−0.10	3
Double DQN	27	750,000	25.85	2,331	12.40	1,228	1.59	128	0.00	1
Double DQN	27	1,000,000*	28.56	2,404	13.69	1,274	1.74	145	−0.06	4
Double DQN	27	final	28.59	2,413	13.75	1,280	1.69	146	−0.06	4
FQI	7	–	29.53	2,476	9.80	1,756	0.82	197	−0.01	7
FQI	17	–*	29.14	2,440	9.78	1,653	1.12	198	0.21	7
FQI	27	–	29.46	2,446	8.88	1,669	0.73	199	−0.01	7
Contextual bandit	7	–	13.17	2,860	2.75	619	−0.01	36	−0.01	1
Contextual bandit	17	–	13.47	2,723	2.72	589	−0.09	44	−0.02	2
Contextual bandit	27	–*	13.22	2,924	2.48	669	0.01	41	−0.01	1

E Every PPO run on the test days

Table 17: Each PPO run’s best checkpoint under its group’s selection rule, on test A at the group’s latency (fast replay): net P&L in US dollars and trades. Run labels give the seed and the entropy coefficient. The selected runs are seed 27, entropy 0.01, in both groups.

ms	Run	0.5 bps		1 bps		2 bps		3.5 bps	
		Net	Tr.	Net	Tr.	Net	Tr.	Net	Tr.
150	seed 7, 0.01	107.50	5,344	64.77	4,669	13.08	1,523	1.53	189
150	seed 7, 0.03	105.54	5,603	64.05	4,931	12.01	1,224	1.56	129
150	seed 17, 0.01	105.27	5,408	62.81	4,435	12.11	1,108	1.68	149
150	seed 17, 0.03	103.07	5,629	63.10	4,789	12.27	994	1.81	99
150	seed 27, 0.01	107.82	5,340	66.07	4,556	13.51	978	1.71	85
150	seed 27, 0.03	103.54	5,467	61.82	4,296	13.20	1,130	1.70	146
450	seed 7, 0.01	75.99	5,133	37.22	3,724	3.22	490	−0.13	34
450	seed 17, 0.01	75.02	4,925	34.95	3,491	3.52	349	0.22	25
450	seed 27, 0.01	76.91	5,056	37.91	3,813	2.21	628	0.05	45

All six 150 ms runs and all three 450 ms runs made money at 0.5, 1 and 2 bps; one 450 ms run lost \$0.13 at 3.5 bps. The spread between runs (for example \$75.02 to \$76.91 at 0.5 bps and 450 ms) is small compared with the differences between algorithms.

F Per-day test results

Table 18: Primary test by day (450 ms, unchanged simulator): net P&L in US dollars with trades in parentheses.

Fee	Policy	29 Sep	30 Sep	1 Oct	2 Oct
0.5	PPO	13.42 (946)	20.97 (1,345)	20.21 (1,274)	22.31 (1,491)
	Double DQN	8.56 (713)	15.83 (1,180)	14.17 (1,032)	16.06 (1,293)
	FQI	10.51 (818)	12.69 (1,027)	15.52 (1,020)	16.65 (1,159)
	Contextual bandit	3.90 (932)	7.79 (1,374)	6.98 (1,243)	6.85 (1,532)
	Best rule (convergence exit)	13.65 (939)	19.76 (1,345)	20.33 (1,236)	23.03 (1,452)
	Raw-gap rule (version 1)	2.94 (193)	7.06 (810)	4.81 (330)	9.12 (725)
1.0	PPO	6.00 (701)	10.92 (1,033)	10.00 (917)	10.99 (1,162)
	Double DQN	3.24 (337)	7.79 (675)	6.21 (509)	8.05 (665)
	FQI	3.90 (514)	3.77 (685)	6.29 (664)	6.37 (822)
	Contextual bandit	0.39 (168)	1.66 (403)	1.67 (265)	0.79 (357)
	Best rule (convergence exit)	5.80 (939)	8.46 (1,345)	9.95 (1,236)	10.59 (1,452)
	Raw-gap rule (version 1)	0.16 (6)	1.88 (78)	0.33 (12)	1.41 (62)
2.0	PPO	−0.15 (59)	0.78 (239)	0.71 (122)	0.88 (208)
	Double DQN	0.03 (16)	1.53 (96)	0.85 (52)	0.76 (76)
	FQI	−0.11 (21)	−0.35 (152)	0.97 (63)	0.29 (113)
	Contextual bandit	−0.02 (5)	0.45 (30)	0.20 (14)	0.13 (17)
	Best rule (convergence exit)	0.18 (15)	0.85 (88)	0.45 (41)	0.47 (66)
	Raw-gap rule (version 1)	0.04 (1)	0.41 (6)	0.02 (1)	0.20 (7)
3.5	PPO	−0.01 (2)	0.08 (19)	0.01 (6)	−0.04 (18)
	Double DQN	0.01 (2)	0.13 (5)	0.02 (1)	−0.15 (4)
	FQI	−0.10 (2)	0.17 (13)	0.01 (4)	−0.14 (8)
	Contextual bandit	0.02 (1)	0.05 (3)	0.00 (0)	−0.01 (1)
	Best rule (convergence exit)	−0.06 (2)	0.11 (12)	−0.03 (6)	0.02 (10)
	Raw-gap rule (version 1)	0.00 (0)	0.00 (0)	0.00 (0)	0.00 (0)

G Full stress grid, 450 ms group

Table 19: Frozen 450 ms choices on test A under each stress (fast replay): net P&L in US dollars and trades. Fee 4.5 bps (extrapolation): PPO $-\$0.10$ (13 trades), best rule $-\$0.46$ (30), fixed-hold rule $-\$0.15$ (11), raw-gap rule $\$0.00$ (0).

Policy	0.5 bps		1 bps		2 bps		3.5 bps	
	Net	Tr.	Net	Tr.	Net	Tr.	Net	Tr.
<i>fill 150 ms</i>								
PPO	107.51	5,117	63.36	3,894	10.34	666	1.50	47
Best rule (convergence exit)	108.29	5,047	65.69	5,047	6.13	231	1.22	33
De-meaned rule, fixed hold	51.67	4,961	11.90	530	2.19	31	1.15	11
Raw-gap rule (version 1)	36.18	2,090	7.10	167	1.60	16	0.22	1
<i>fill 300 ms</i>								
PPO	91.55	5,085	50.27	3,853	7.09	641	1.14	49
Best rule (convergence exit)	91.53	5,011	49.22	5,011	4.34	215	0.92	34
De-meaned rule, fixed hold	40.41	4,912	8.50	506	1.89	33	0.90	14
Raw-gap rule (version 1)	29.76	2,043	5.71	162	1.51	17	0.23	1
<i>fill 450 ms (primary)</i>								
PPO	76.91	5,056	37.91	3,813	2.21	628	0.05	45
Best rule (convergence exit)	76.77	4,972	34.80	4,972	1.94	210	0.05	30
De-meaned rule, fixed hold	28.95	4,886	4.34	508	0.61	28	0.04	11
Raw-gap rule (version 1)	23.92	2,058	3.78	158	0.67	15	0.00	0
<i>fill 600 ms</i>								
PPO	58.96	5,032	23.88	3,783	-0.61	624	0.19	45
Best rule (convergence exit)	58.24	4,966	16.32	4,966	0.91	212	0.06	31
De-meaned rule, fixed hold	16.69	4,893	2.54	504	1.19	28	0.45	11
Raw-gap rule (version 1)	17.65	2,053	3.66	155	1.08	17	0.15	1
<i>spread $\times 1.5$</i>								
PPO	71.11	5,056	33.42	3,813	1.21	627	-0.05	45
Best rule (convergence exit)	70.86	4,972	28.89	4,972	1.48	210	-0.03	30
De-meaned rule, fixed hold	23.85	4,886	3.49	508	0.55	28	0.01	11
Raw-gap rule (version 1)	21.54	2,058	3.46	158	0.62	15	0.00	0
<i>HL-clock fill</i>								
PPO	27.05	5,049	-4.62	3,806	-10.07	621	-2.21	44
Best rule (convergence exit)	26.11	4,974	-15.88	4,974	-3.74	212	-1.37	29
De-meaned rule, fixed hold	-4.09	4,887	-3.54	509	-0.37	27	-0.49	11
Raw-gap rule (version 1)	4.23	2,056	-0.27	157	-0.18	15	0.00	0

H Additional training curves

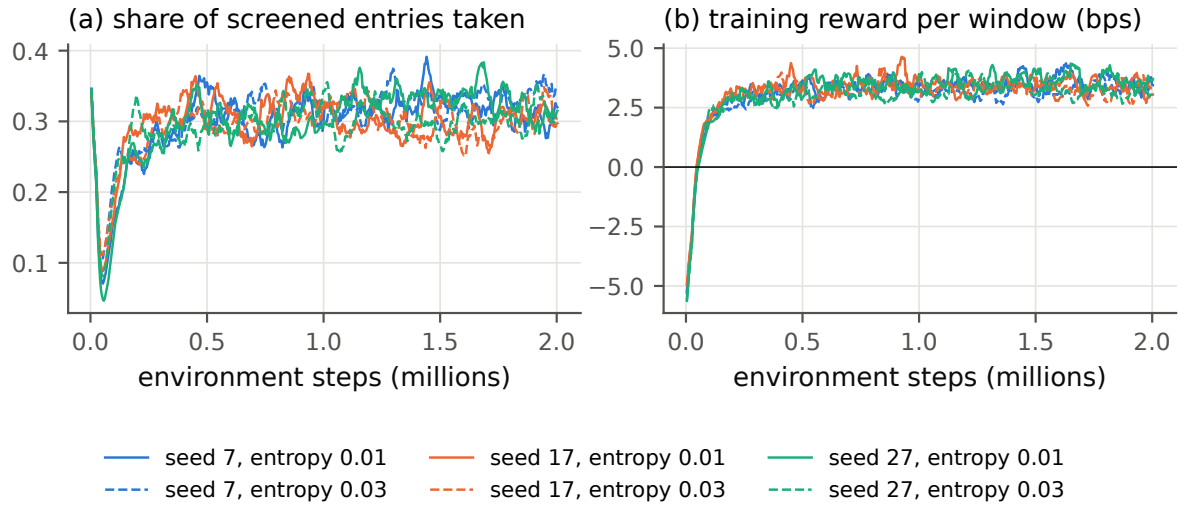


Figure 11: The six 150 ms PPO runs: share of screened entries taken and training reward per window, smoothed over ten log points.

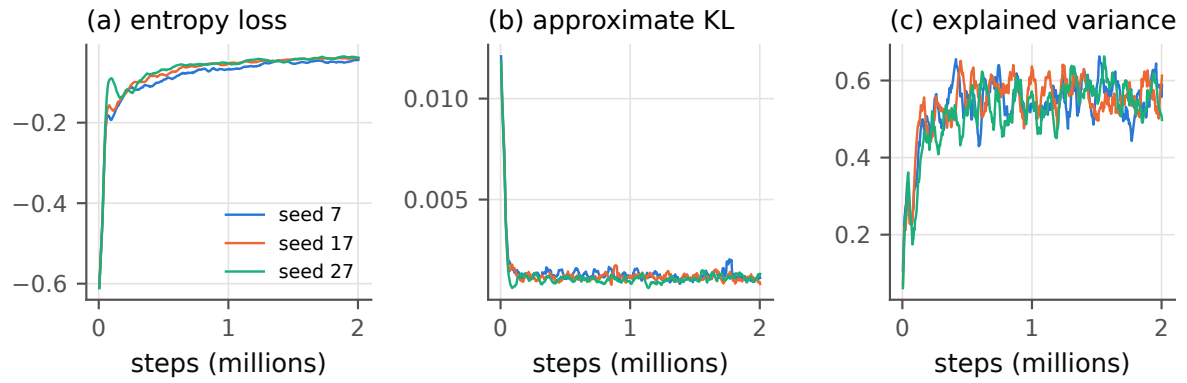


Figure 12: PPO optimisation statistics for the three 450 ms runs, logged per update and smoothed: entropy loss (the negative policy entropy), approximate KL divergence between successive policies, and the fraction of return variance explained by the value function.

I Feed timing

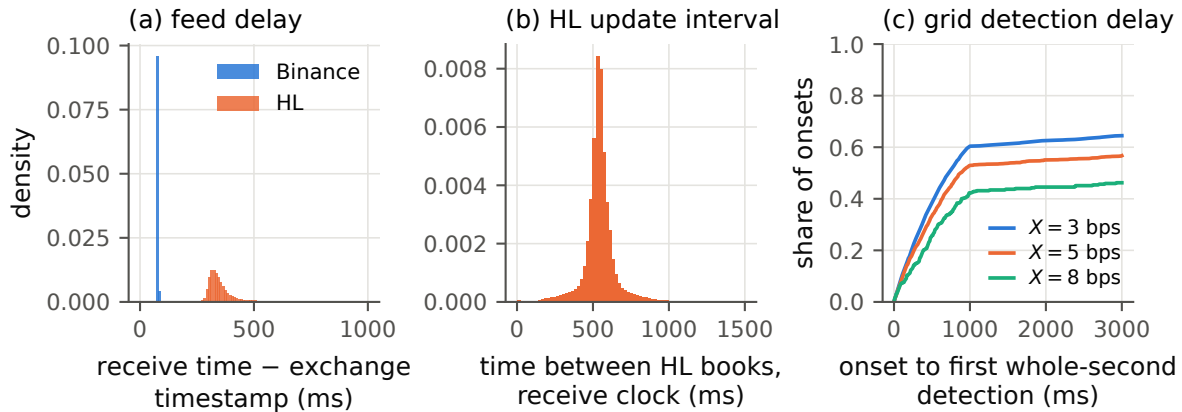


Figure 13: Six training days. (a) Receive time minus exchange timestamp per venue (clock offsets included); Binance median 75.8 ms, HL median 338.3 ms. (b) Time between HL book updates on the receive clock (median 540 ms). (c) Share of de-meaned-gap onsets at threshold X detected by the one-second grid within a given delay. Recomputed read-only from the recorded data with the functions of `analysis/subsecond_edge.py`; the percentiles match its stored output.

J Deployment export

`latency_arb/v2/export_policy.py` wrote the 150 ms group’s selected actor (seed 27, entropy 0.01, final checkpoint) as ONNX, as raw float32 weights and as a feature specification in `runs/v2-leadlag-ppo/export/`. The network maps the 20 scaled and clipped features through two 64-unit tanh layers to two logits, and the action is the argmax (flat: skip or enter in the sign of the de-meaned gap; holding: hold or exit). On 24,736 checked observations the ONNX model and Stable-Baselines3 chose the same action every time. The primary 450 ms policy has not been exported.